1 08.28.23 - Lecture #1 - Introduction

Riley Thornton (rthornto@andrew.cmu.edu)
Office: Wean 7101
Office Hours: 11:30 - 1:30 Mondays

Definition 1.1. Informally, a **vector space** is a structure where we can add and scale.

Definition 1.2. If V, W are vector spaces, $f : V \to W$ is a **function**.

Definition 1.3. We say f is **linear** if $f(x+y) = f(x) + f(y)$ and $f(\lambda x) = \lambda f(x)$.

Example 1.4. What to make for breakfast?

V = lists of recipes, ex. 3 waffles, 2 pancakes, 1 omelette
W = ingredient amounts, ex. 2 c. flour, 1 c. milk, 5 eggs

$f(\vec{v})$ = list of ingredients needed

How to compute f?

1 waffle takes 1 c. flour, 1/2 c. milk, 1 egg
1 pancake takes 1 c. flour, 1 c. milk, 1 egg
1 omelette takes 2 eggs

$f(x \text{ waffles}, y \text{ pancakes}, z \text{ omelettes}) = xf(1 \text{ waffle}) + yf(1 \text{ pancake}) + zf(1 \text{ omelette}) = (x+y) \text{ flour } + (\frac{1}{2}x + y) \text{ milk } + (x + y + 2z) \text{ eggs}$

This example now leads us to ask the question:

Question. Given $\vec{y} \in W$, is there $\vec{x} \in V$ with $f(\vec{x}) = \vec{y}$? Is there always a solution?

Example 1.5. Measuring Gravity

V = space of polynomials in one variable

$W = \mathbb{R}^n$

Specify $t_1, t_2, \cdots, t_n \in \mathbb{R}$ then $f(p) = \begin{bmatrix} p(t_1) \\ p(t_2) \\ \vdots \\ p(t_n) \end{bmatrix}$

From elementary physics knowledge, we know that

$$-\frac{1}{2}gt^2 + v_0 t + d_0 = d$$

If we didn't know any of this data, we could find it by tossing a ball upward and

> measuring at times $t_0, t_1, \cdots, t_n$, and we will get $f(p)$; then we only need to find $p \in V$ with $f(p) = \vec{v}$.
>
> But, there could be an error in measurement and there could be no solution; we'll have to find an approximate solution.

2 08.30.23 - Lecture #2 - Formalizing Vector Spaces

How do we formalize the above notion of a vector space?

Definition 2.1. Fix $\mathbb{F}$ a field (i.e. $\mathbb{F} = \mathbb{Q}, \mathbb{R}, \mathbb{C}$ for this class).

A vector space over $\mathbb{F}$ is a structure $(V, \oplus, \odot)$ with the operations

$$\oplus : V^2 \to V$$

$$\odot : \mathbb{F} \times V \to V$$

that satisfy the following properties:

1. **Commutativity:** $x \oplus y = y \oplus x$
2. **Associativity:** $x \oplus (y \oplus z) = (x \oplus y) \oplus z$
3. **Additive identity:** there exists some $z \in V$ such that $z \oplus x = x$
4. **Inverse:** for all $x, z \in V$, there exists some $y \in V$ such that $(y \oplus x) \oplus z = z$
5. **Left distributivity:** $\alpha \odot (x \oplus y) = (\alpha \odot x) \oplus (\alpha \odot y)$
6. **Right distributivity:** $(\alpha + \beta) \odot x = (\alpha \odot x) \oplus (\beta \odot x)$
7. **Multiplicative identity:** $1 \odot x = x$
8. **Associativity of multiplication:** $(\alpha \cdot \beta) \odot x = \alpha \odot (\beta \odot x)$

The following are some examples of vector spaces:

Example 2.2. $(\mathbb{F}, +, \cdot)$ is a vector space over $\mathbb{F}$

Example 2.3. $\mathbb{F}^n = \{ \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \mid x_1, \cdots, x_n \in \mathbb{F}\}$ with the usual operations.

Example 2.4. For a set A, $\mathbb{F}^A = \{f \mid f \text{ is a function from } A \text{ to } \mathbb{F}\}$ with operations

$$(f \oplus g)(x) = f(x) + g(x)$$

$$(\alpha \odot f)(x) = \alpha f(x).$$

The additive identity is $\vec{0}(x) = 0$ for all x.

Example 2.5. $P(\mathbb{F}) = \{p \in \mathbb{F}^{\mathbb{F}} : p \text{ is a polynomial}\}$

$\exists a_1, \cdots, a_n \in \mathbb{F}$ $p(x) = a_n x^{n-1} + \cdots + a_2 x + a_1$ with inherited operations from $\mathbb{F}^A$

Fact: If $p \in P(\mathbb{R})$ and $p(x) = 0$ for all x and $p = \sum_{i=0}^{n} a_i x^i$, then $a_i = 0$ for all i.

Example 2.6. $\{p \in P(\mathbb{F}) \mid \deg(p) = n\}$ with the usual operations is NOT a vector space because $\oplus$ isn't even defined for this set since $x, -x \in \{p \in P(\mathbb{F}) \mid \deg(p) = 1\}$ but $x \oplus (-x)$ is NOT.

Example 2.7. $P_n(\mathbb{F}) = \{p \in P(\mathbb{F}) \mid \deg(p) \leq n\}$ is a vector space with the inherited operations

We can prove some properties about vector spaces from these basic properties:

Theorem 2.8. The additive identity in any vector space is unique, i.e. if $\vec{x}, \vec{y} \in V$ and $(\forall \vec{z} \in V) \vec{x} \oplus \vec{z} = \vec{y} \oplus \vec{z} = \vec{z}$ then $\vec{x} = \vec{y}$

Proof. Fix $\vec{x}, \vec{y} \in V$ such that $\forall \vec{z}$,

$$\vec{x} \oplus \vec{z} = \vec{y} \oplus \vec{z} = \vec{z},$$

then in particular

$$\vec{x} \oplus \vec{y} = \vec{y} \oplus \vec{x} = \vec{x}$$

and $\vec{x} \oplus \vec{y} = \vec{y}$, so $\vec{x} = \vec{y}$. □

Now, we can denote the additive identity in V as $\vec{0_v}$.

Theorem 2.9. For any vector $\vec{v} \in V$, $0 \odot \vec{v} = \vec{0}_v$

Proof. Fix $\vec{v} \in V$ and consider

$$(1 + 0) \odot \vec{v} = (1 \odot \vec{v}) \oplus (0 \odot \vec{v}) = \vec{v} \oplus 0 \cdot \vec{v}$$

$$(1 + 0) \odot \vec{v} = 1 \odot \vec{v} = \vec{v}$$

So $\vec{v} = \vec{v} \oplus 0 \odot \vec{v}$.

Let $\vec{z}$ be an additive inverse for $\vec{v}$, so

$$\begin{aligned} \vec{0}_v &= \vec{z} \oplus \vec{v} \\ &= \vec{z} \oplus \vec{v} \oplus 0 \odot \vec{v} \\ &= \vec{0}_v \oplus 0 \odot \vec{v} \\ &= 0 \odot \vec{v} \end{aligned}$$

□

In a similar way, we can prove:

1. Each vector has a unique additive inverse
2. For any $\vec{v} \in V$, $(-1) \odot \vec{v}$ is the additive inverse of $\vec{v}$
3. If $\vec{v} \neq \vec{0}_v$ and $\alpha, \beta \in \mathbb{F}$ with $\alpha \neq \beta$, then $\alpha \odot \vec{v} \neq \beta \odot \vec{v}$.

3 09.01.23 - Lecture #3 - Linear Independence

Definition 3.1. Given a list of vectors from V, $\mathcal{L} = (\vec{v}_1, \cdots, \vec{v}_n)$.

A **linear combination** of $\mathcal{L}$ is an expression

$$\alpha_1\vec{v}_1 + \alpha_2\vec{v}_2 + \cdots + \alpha_n\vec{v}_n.$$

We call α_i the **coefficient** of $\vec{v}_i$.

For example,

$$\frac{1}{2}\begin{bmatrix}1\\2\\4\end{bmatrix} + \frac{3}{2}\begin{bmatrix}1\\0\\0\end{bmatrix} = \begin{bmatrix}2\\1\\2\end{bmatrix}$$

is a linear combination of $\left(\begin{bmatrix}1\\2\\4\end{bmatrix}, \begin{bmatrix}1\\0\\0\end{bmatrix}\right)$

Definition 3.2. Fix $\mathcal{L} = (v_1, \cdots, v_n)$. Then the **span** of $\mathcal{L}$ is defined as $\operatorname{span}(\mathcal{L}) = \{\alpha_1\vec{v}_1 + \cdots + \alpha_n\vec{v}_n \mid \alpha_1, \cdots, \alpha_n \in \mathbb{F}\}$.

The empty list is $()$; $\operatorname{span}() = \{0\}$ by convention.

Example 3.3. $\operatorname{span}(\begin{bmatrix}1\\1\end{bmatrix}) = \{\begin{bmatrix}a\\a\end{bmatrix} \mid a \in \mathbb{R}\}$

$\operatorname{span}(\begin{bmatrix}1\\1\end{bmatrix}, \begin{bmatrix}1\\-1\end{bmatrix} = \mathbb{R}^2$

Proposition. $\{p \in P(\mathbb{R}) \mid p(x) = p(-x)\} = \operatorname{span}(\{x^n \mid n \in 2\mathbb{N}\}$

Proof. We first prove the forward direction.

Suppose $\vec{p} \in \operatorname{span}(\{x^n \mid n \in 2\mathbb{N})\}$. This means

$$p(x) = a_0 + a_2x^2 + a_4x^4 + \cdots + a_{2n}x^{2n} = \sum_{i=0}^{n} a_{2i}x^{2i}$$

for some $a_0, a_2, \cdots, a_{2n} \in \mathbb{R}$.

So

$$p(-x) = \sum_{i=0}^{n} a_{2i}(-x)^{2i} = \sum_{i=0}^{n} a_{2i}(-1)^{2i}x^{2i} = \sum_{i=0}^{n} a_{2i}x^{2i}$$

We now prove the converse.

Suppose $p(x) = p(-x)$, say $p(x) = \sum_{i=0}^{n} a_ix^i$. Then

$$\vec{0} = p(x) - p(-x) = \sum_{i=0}^{n} a_ix^i - \sum_{i=0}^{n} a_i(-x)^i$$

$$= \sum_{\substack{i=0 \\ i \text{ odd}}}^{n} 2a_i x^i + \sum_{\substack{i=0 \\ i \text{ even}}}^{n} (a_i - a_i)x^i.$$

And we have that if $\sum_{i=0}^{n} b_i x^i = \vec{0}$, then $b_i = 0$ for all i.

So $2a_i = 0$ for all odd i, i.e. $p(x) = \sum_{\substack{i=0 \\ i \text{ even}}}^{n} a_i x^i$

$\square$

In general, we think of a linear combination of $\mathcal{L}$,

$$\alpha_1 \vec{v}_1 + \cdots + \alpha_n \vec{v}_n = \vec{v}$$

as giving a **representation** of $\vec{v}$ in terms of $\mathcal{L}$. Instead of working with sums, we would like to say that working with v is the same as working with a vector of coefficients in a particular representation. However, this requires representations to be unique.

For example, if $\mathcal{L} = \left(\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 3 \\ 3 \\ 6 \end{bmatrix} \right)$, then

$$\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} - \begin{bmatrix} 3 \\ 3 \\ 6 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} = 0 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 3 \\ 3 \\ 6 \end{bmatrix}$$

We find that our list is redundant because the third vector is in the span of the first two.

Definition 3.4. A list of vectors $\mathcal{L}$ is an **ordered basis** if every vector has a unique representation in the in terms of $\mathcal{L}$. Equivalently, $\mathcal{L}$ is an **ordered basis** if $\mathcal{L}$ is both spanning and linearly independent.

Example 3.5. The *standard ordered basis* for $\mathbb{F}^n$ is defined to be $\mathcal{L}_{st} = (e_1, e_2, \cdots, e_n)$, where

$$e_i = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix}$$

Definition 3.6. A list $\mathcal{L}$ is **spanning** or **generating** for V if $\text{span}(\mathcal{L}) = V$, i.e. every vector in V has at least one representation in terms of $\mathcal{L}$.

Definition 3.7. $\mathcal{L}$ is **linearly independent** if every vector has at most one representation in terms of $\mathcal{L}$.

Proposition. A list $\mathcal{L}$ is linearly independent iff the zero vector has a unique representation in terms of $\mathcal{L}$.

Exercise 3.1. Let V be a vector space over K, $A \subseteq V$. Prove that $\text{span}(A) = A$ if and only iff for every $\alpha \in K$, $v, w \in A$, $\alpha v + w \in A$.

Exercise 3.2. Span is a closure operator.

1. $A \subseteq \text{span}(A)$
2. $A \subseteq B \implies \text{span}(A) \subseteq \text{span}(B)$
3. $\text{span}(\text{span}(A)) = \text{span}(A)$

Exercise 3.3. Claim: $\text{span}(A)$ is the smallest set $B \supseteq A$ such that B is closed under $+$ and $\cdot$.

If C is any other set such that $C \supseteq A$, C closed under $+, \cdot$, then $\text{span}(A) \subseteq C$

4 09.06.23 - Lecture #4 - Basis Examples

What happens if we'd like to work with (unordered) sets, instead of lists?

Definition 4.1. A set $A \subseteq V$ is **spanning** or **generating** for V if $\operatorname{span}(A) = V$.

Definition 4.2. A set A is **linearly independent** if every list of distinct elements from A is linearly independent. Note that if $A = \{\vec{v}_1, \cdots, \vec{v}_n\}$ with $\vec{v}_i$'s distinct, it suffices to check $(\vec{v}_1, \cdots, \vec{v}_n)$.

Example 4.3. The standard basis $B_{st} = \{e_1, \cdots, e_n\}$ is an (unordered) basis for $\mathbb{F}^n$.

Example 4.4. The set $B = \{x^i \mid i \in \mathbb{N}\}$ is a basis for $P(\mathbb{F})$.

Example 4.5. Let $V = \{\begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 : x + y + z = 0\}$ with the usual operations, then

$$\left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \right\}$$

is a basis for V.

Proof. We can check that this is spanning and linearly independent.

Spanning: Fix $\begin{bmatrix} x \\ y \\ z \end{bmatrix} \in V$ with $x + y + z = 0$, so $x = -y - z$. Then

$$y \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} - z \begin{bmatrix} 1 \\ 0 \\ = 1 \end{bmatrix} = \begin{bmatrix} -y - z \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Linearly independent: Suppose

$$\alpha \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} \alpha + \beta \\ -\alpha \\ -\beta \end{bmatrix},$$

then $-\alpha = -\beta = 0$, so $\alpha = \beta = 0$. □

Proposition 4.6. $\{\frac{1}{2}(x-1)(x-2), -1(x-2)x, \frac{1}{2}(x-1)x\}$ is a basis for $P_2(\mathbb{F})$

Theorem 4.7. $A \subseteq V$ is linearly independent if and only if it has no "redundanct vectors", i.e.

$$\forall v \in A, v \notin \operatorname{span}(A \setminus \{v\})$$

Proof. Prove both directions by contraposition.

We first prove the forward direction. Suppose $v \in A$ is redundant, i.e. there are $u_1, \cdots, u_n \in A$ distinct $u_i \neq v$ for all i, such that $v = \alpha_1 u_1 + \cdots + \alpha_n u_n$. Then v has 2 representations in terms of $v, u_1, \cdots, u_n)$, so this list is not linearly independent, so A is not

We now prove the reverse. Suppose A is not linearly independent, and we'll show that it has a redundant vector. There are $v_1, \cdots, v_n \in A$ distinct such that $\alpha_1 v_1 + \cdots + \alpha_n v_n = \vec{0}$ with $\alpha_i \neq 0$ for some i, then $-v_i = \frac{\alpha_1}{\alpha_i} v_1 + \cdots + \frac{\alpha_n}{\alpha_i} v_n \in \operatorname{span}(A \setminus \{v_i\})$. $\square$

Theorem 4.8. If V has a finite generating set, then V has a basis. In fact, any generating set contains a basis.

Proof. Pick any finite generating set A of V by hypothesis. Suppose A is not linearly independent, then there exists some redundant vector $v \in A$. This implies that $v \in \operatorname{span}(A \setminus \{v\})$, so $A \setminus \{v\}$ is a generating set of V. Repeat this process by induction on $|A \setminus \{v\}|$ until it becomes linearly independent. $\square$

Exercise 4.1. If V has a finite generating set, any generating set contains a finite generating set.

Exercise 4.2. $\mathbb{R}[x]$ has no finite spanning set.

Exercise 4.3. For a vector space V, a set $S \subseteq V$ is linearly independent if and only if all finite subsets of S are independent.

5 09.08.23 - Lecture #5 - Dimension

We might notice that many of the spaces that we've looked at resemble $\mathbb{F}^n$, and we might wonder if there is a way to define a "dimension" on these spaces to relate it to some given $\mathbb{F}^n$.

Theorem 5.1 (Dimension Theorem). If A, B are finite bases for a vector space V, then

$$|B| = |A|.$$

Proof. Suppose for the sake of contradiction that, without loss of generality, $|A| > |B| = n$. By the below lemma, we have

$$v_1, \cdots, v_n \in A$$

with $A' = (A \setminus \{v_1, \cdots, v_n\} \cup B$ a basis. Since $|A| > n$, A' contains some vector $v \notin B$. But then A' is not linearly independent because $v \in \operatorname{span}(B)$. □

Lemma 5.2. For all $1 \leq i \leq n$, there are $u_1, u_2, \cdots, u_i \in B$ distinct and $v_1, v_2, \cdots, v_i \in A$ distinct with $(A \setminus \{v_1, \cdots, v_i\}) \cup \{u_1, \cdots, u_i\}$ still a basis.

Proof. By induction on i.

Base case: (i=1) Fix any $a_1 \in B$.

Because A is spanning, $u_1 = \alpha_1 w_1 + \alpha_2 w_2 + \cdots + \alpha_m w_m$ for some $w_1, \cdots, w_m \in A$ distinct and $\alpha_1, \cdots, \alpha_m \in \mathbb{F}$ nonzero.

Let $v_1 = w_1$. We want to show that $(A \setminus \{w_1\}) \cup \{u_1\}$ is spanning and linearly independent.

Spanning: It suffices to show that $w_1 \in \operatorname{span}(A \setminus \{w_1\}) \cup \{u_1\}$, $w_1 = \frac{1}{\alpha_1} u - \frac{\alpha_2}{\alpha_1} w_2 - \cdots - \frac{\alpha_m}{\alpha_1} w_m$.

Linearly independent: (HW) if S is linearly independent and $v \notin \operatorname{span}(S)$, then $S \cup \{v\}$ is linearly independent. So it suffices to show that $u_1 \notin \operatorname{span}(A \setminus \{w_1\}$. If it were, u_1 would have two representations in terms of distinct vectors from A (1 rep with w_1 and 1 withot). This can't happen since A is linearly independent.

Induction step: Suppose $i < n$ and we have $v_1, \cdots, v_i \in A$ distinct and $u_1, \cdots, u_i \in B$ distinct with $(A \setminus \{v_1, \cdots, v_i\}) \cup \{u_1, \cdots, u_n\}$ a basis.

Consider any $u_{i+1} \in B \setminus \{u_1, \cdots, u_i\}$. Then $u_{i+1} \in \operatorname{span}(A')$ and because B is linearly independent, $u_{i+1} \notin \operatorname{span}(u_1, \cdots, u_i)$. So for some $\alpha_1, \cdots, \alpha_m \in \mathbb{F}$ nonzero, $u_{i+1} = \alpha_1 w_2 + \alpha_2 w_2 + \cdots + \alpha_m w_m \in A$ with $w_1, \cdots, w_m$ distinct and some $w_j \notin \{u_1, \cdots, u_i\}$. Then by the same argument as in the base case, $(A' \setminus \{w_j\}) \cup \{u_{i+1}\}$ is a basis. □

Definition 5.3. The **dimension** of a vector space V given some basis B is

$$\dim(V) = |B|.$$

Example 5.4. $\dim(\mathbb{F}^n) = n$

Note that when $n = 0$, the empty set is the basis for $\mathbb{F}^0$.

Example 5.5. $\dim(P_n(\mathbb{F})) = n + 1$

Example 5.6. $V = \{\begin{bmatrix} x \\ y \\ z \end{bmatrix} : x - y + z = 0\}$ with operations inherited from $\mathbb{R}^3$ has dimension 2.

To find a basis, we note that $y = x + z$, so

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ x+z \\ z \end{bmatrix} = x \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

Theorem 5.7. For $B \subseteq V$, V a vector space, $\dim(V) = n < \infty$.

Then any 2 of the following properties imply B is a basis (and also implies the 3rd property).

1. B is linearly independent
2. B is spanning
3. $|B| = n$

Proof. $(1 \ \& \ 2) \to 3$ is the dimension theorem

$(2 \ \& \ 3) \to 1:$ we showed if B is spanning, there is $B' \subseteq B$ with B' a basis. If $|B| = n = |B'|$, so $B = B'$

$(1 \ \& \ 3) \to 2:$ (HW) If B is linearly independent, $B \subseteq B'$ for some basis B'.

So $|B| = n = |B'|$ and $B = B'$. $\square$

6 09.11.23 - Lecture #6 - Linear functions

Definition 6.1. For V, W vector spaces, $f : V \to W$ is **linear** if $(\forall v, u \in V, \alpha \in \mathbb{F})$,

$$f(u \oplus v) = f(u) \oplus f(v)$$

$$f(\alpha v) = \alpha f(v)$$

Example 6.2. The derivative map.

$V = P_n(\mathbb{R}), W =_{n-1} (\mathbb{R})$

$$\frac{\mathrm{d}}{\mathrm{d}x} : V \to W$$

$$\frac{\mathrm{d}}{\mathrm{d}x} p = (\frac{\mathrm{d}}{\mathrm{d}x} p(x))$$

Then

$$\frac{\mathrm{d}}{\mathrm{d}x}(a_0 + a_1 x + \cdots + a_n x^n) = a_1 + 2a_2 x + \cdots + n a_n x^{n-1}$$

Example 6.3. The integral map.

$$\int : W \to V$$

$$\int p = \int_0^x p(t)\,\mathrm{d}t$$

Proposition 6.4. If $f, g : V \to W$ are linear and $h : W \to X$, then $(f + g), \alpha f$ for scalar α, and $h \circ f$ are also linear

Lemma 6.5. If $f, g : V \to W$ are linear, S is spanning for V, and for all $v \in S$, $f(v) = g(v)$, then $f = g$.

Proof. Fix $v \in V$. Since S is spanning, $v = \alpha_1 v_1 + \cdots + \alpha_n v_n$ with $v_1, \cdots, v_n \in S$.

Then

$$\begin{aligned} f(v) = f(\alpha_1 v_1 + \cdots + \alpha_n v_n &= \alpha_1 f(v_1) + \cdots + \alpha_n f(v_m) \\ &= \alpha_1 (g_1 \alpha_1) + \cdots + \alpha_n g(v_n) \\ &= g(\alpha_1 v_1 + \cdots + \alpha_n v_n) = g(v) \end{aligned}$$

Thus for any $v \in V$, $f(v) = g(v)$, meaning $f = g$. $\square$

Lemma 6.6. If $S \subseteq V$ is linearly independent and $f : S \to W$ is any function, then there is $f' : V \to W$ which is linear and $\forall v \in S$, we have that $f'(v) = f(v)$.

Proof. By HW there is some basis $B \supseteq S$. Set $g : B \to W$

$$g(v) = \begin{cases} f(v) & \text{if } v \in S \\ 0 & \text{otherwise} \end{cases}$$

Define $f' : V \to W$ by

$$f'(v) = \alpha_i g(v_1) + \cdots + \alpha_n g(v_n)$$

if $v = \alpha_1 v_1 + \cdots + \alpha_n v_n$ with $v_1, \cdots, v_n \in B$ distinct. It is easy to check that f' is linear. $\square$

Theorem 6.7. If B is a basis for V, then for any $f : B \to W$ there is a unique linear extension $f' : V \to W$ with $f'(v) = f(w)$ for $v \in B$.

Definition 6.8. If $\mathcal{L} = (v_1, \cdots, v_n)$ is an ordered basis for V, then the **coordinate map** for $\mathcal{L}$ is the unique linear function extending the map $v_1 \mapsto e_1 \in \mathbb{F}^n$.

So

$$\left[v\right]_{\mathcal{L}} = \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix}$$

if

$$v = \alpha v_1 + \cdots + \alpha_n v_n$$

If $V = \mathbb{F}^n$ and $\mathcal{L} = \{e_1, e_2, \cdots, e_n\}$ then $\left[v\right]_{\mathcal{L}} = v$.

If $V = P_n(\mathbb{F})$ and $\mathcal{L} = \{1, x, x^2, \cdots, x^n\}$ then $\left[P\right]_{\mathcal{L}} =$ coefficients of list in ascending order.

Definition 6.9. An **isomorphism** between vector spaces is a linear function $f : V \to W$ with a linear inverse $f^{-1} : W \to V$ i.e. $f \circ f^{-1} = \mathrm{id}_w$ and $f^{-1} \circ f = \mathrm{id}_v$

Proposition 6.10. The coordinate map is an isomorphism.

7 09.13.23 - Lecture #7 - Coordinate Maps

We begin with an example of computing coordinate maps:

Exercise 7.1. Show that

$$\mathcal{L} = (\begin{bmatrix}1\\2\\3\end{bmatrix}, \begin{bmatrix}0\\1\\2\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix})$$

is an ordered basis for $\mathbb{R}^3$ and compute the associated coordinate map

Proof. To show that $\mathcal{L}$ is a basis, it suffices to show that $\mathcal{L}$ linearly independent.

Suppose $\alpha_1 \begin{bmatrix}1\\2\\3\end{bmatrix} + \alpha_2 \begin{bmatrix}0\\1\\2\end{bmatrix} + \alpha_3 \begin{bmatrix}0\\0\\1\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix}$, then $\alpha_1 = 0$ and $2\alpha_1 + \alpha_2 = 0 \implies 0 + \alpha_2 = 0$ such that $\alpha_2 = 0$. Similarly, $3\alpha_1 + 2\alpha_2 + \alpha_3 = 0 \implies 0 + 0 + \alpha_3 = 0$ so $\alpha_3 = 0$

Now we want to compute $[v]_{\mathcal{L}}$ for $v \in \mathbb{R}^3$. Suppose $\begin{bmatrix}x\\y\\z\end{bmatrix} \in \mathbb{R}^3$, then first solve

$$\alpha_1 \begin{bmatrix}1\\2\\3\end{bmatrix} + \alpha_2 \begin{bmatrix}0\\1\\2\end{bmatrix} + \alpha_3 \begin{bmatrix}0\\0\\1\end{bmatrix} = \begin{bmatrix}x\\y\\z\end{bmatrix}.$$

We get that $\alpha_1 = x$, $2\alpha_1 + \alpha_2 = y \implies \alpha_2 = y - 2x$, $3\alpha_1 + 2\alpha_2 + \alpha_3 = z \implies \alpha_3 = z - 2y + x$, so

$$\begin{bmatrix}x\\y\\z\end{bmatrix}_{\mathcal{L}} = \begin{bmatrix}x\\y-2x\\z-2y+x\end{bmatrix}.$$

□

Proposition 7.1. If $V \to W$ is an isomorphism, then

1. For $S \subseteq V$, $v \in V$, $v \in \operatorname{span}(S) \iff f(v) \in \operatorname{span}(f[S])$
2. $B \subseteq V$ is a basis if and only if $f[B]$ is a basis for W
3. $\dim(v) = \dim(w)$

Proposition 7.2. For any ordered basis $\mathcal{L}$, the coordinate map associated to $\mathcal{L}$ is an isomorphism.

Proof. Say $\mathcal{L} = \{v_1, \cdots, v_n\}$ and $[v]_{\mathcal{L}} : V \to \mathbb{F}^n$.

Define $f : \mathbb{F}^n \to V$ such that $f \begin{bmatrix}a_1\\\vdots\\a_n\end{bmatrix} = a_1 v_! + a_2 v_2 + \cdots + a_n v_n$. Then f is a linear inverse

for $[v]_{\mathcal{L}}$. It is easy to show that

$$f([v]_{\mathcal{L}}) = f(\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}) = a_1 v_1 + \cdots + a_n v_n = v.$$

$\square$

Proposition. If V is a finite dimensional vector space, V is isomorphic to $\mathbb{F}^n$ for exactly one value n, namely $n = \dim(V)$

Definition 7.3. A $n \times m$ matrix (rows x columns) over $\mathbb{F}$ is a $n \times m$ rectangle of scalars.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{bmatrix} = (a_{ij} : i \leq n, j \leq n)$$

Definition 7.4 (Matrix multiplication). For a $n \times m$ matrix $A = \begin{bmatrix} \vdots & \vdots & & \vdots \\ v_1 & v_2 & \cdots & v_m \\ \vdots & \vdots & & \vdots \end{bmatrix}$ and some $\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \in \mathbb{F}^n$,

$$A \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} = a_1 v_1 + a_2 v_2 + \cdots + a_n v_n$$

Definition 7.5. For a linear transformation $f : V \to W$, $\mathcal{L} = (v_1, \cdots, v_n), \mathcal{M}$ ordered bases for V, W, $[f]_{\mathcal{L}}^{\mathcal{M}}$ is matrix whose ith column is $[f(v_i)]_{\mathcal{M}}$

8 09.15.23 - Lecture #8 - Matrix Multiplication

The following is an example of matrix multiplication with a vector:

Example 8.1.

$$\begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 2 \\ 3 & 0 & 2 \\ 4 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ 5 \\ 7 \end{bmatrix}$$

Specifically,

$$(Mv)_i = \sum_{j=1}^{n} M_{ij} v_j$$

$$Me_i = 0c_1 + 0c_2 + \cdots + 1c_i + \cdots + 0c_n = c_1,$$

and the ith column of M is equal to Me_i.

Proposition 8.2. For $M \in \mathbb{F}^{m \times n}$ the map $v \mapsto Mv$ from $\mathbb{F}^n$ to $\mathbb{F}^m$ is linear.

Proof. Consider

$$M = \begin{bmatrix} \vdots & \vdots & & \vdots \\ c_1 & c_2 & \cdots & c_n \\ \vdots & \vdots & & \vdots \end{bmatrix} \in \mathbb{F}^{m \times n}$$

$$u = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}, v = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}, \alpha \in \mathbb{F}$$

Then

$$M(\alpha u + v) = (\alpha u_1 + v_1)c_1 + \cdots + (\alpha u_n + v_n)c_n$$

$$= \alpha(u_1 c_1 + u_2 c_2 + \cdots + u_n c_n) + (v_1 c_1 + \cdots + v_n c_n) = \alpha M u + M v$$

□

Definition 8.3 (Change of Basis). For $f : V \to W$ linear, $\mathcal{L}, M$ ordered bases for V, W respectively, $\mathcal{L} = (v_1, \cdots, v_n)$,

$$\left[f\right]_{\mathcal{L}}^{M} = \begin{bmatrix} \vdots & & \vdots \\ [f(v_1)]_M & \cdots & [f(v_n))]_M \\ \vdots & & \vdots \end{bmatrix},$$

so the ith column of $[f]_{\mathcal{L}}^{M}$ is the M-coord vector for f applied to the ith basis vector in $\mathcal{L}$.

Proposition 8.4. For any $f : V \to W$ and ordered bases $\mathcal{L}, M$ for V, W respectively,

$$\left[f\right]_{\mathcal{L}}^{M} \left[v\right]_{\mathcal{L}} = \left[f(v)\right]_{M}$$

Proof. First we check the statement for basis vectors. Say $\mathcal{L} = (v_1, \cdots, v_n)$

$$\left[f\right]_{\mathcal{L}}^{M} \left[v_i\right]_{\mathbb{L}} = \left[f\right]_{\mathcal{L}}^{M} e_i = i\text{th column of } \left[f\right]_{\mathcal{L}}^{M} = \left[f(v_i)\right]_{M}$$

Since the maps $v \mapsto \left[f\right]_{\mathcal{L}}^{M} [v]_{\mathcal{L}}$ and $v \mapsto \left[f(v)\right]_{M}$ are both linear and they agree on a basis, they agree everywhere. □

Definition 8.5 (Matrix Multiplication). For $L \in \mathbb{F}^{l\times m}$ and $M \in \mathbb{F}^{m\times n}$, $LM \in \mathbb{F}^{l\times n}$ is defined as follows:

If $M = \begin{bmatrix} \vdots & & \vdots \\ c_1 & \cdots & c_n \\ \vdots & & \vdots \end{bmatrix}$, then $LM = \begin{bmatrix} \vdots & & \vdots \\ Lc_1 & \cdots & Lc_n \\ \vdots & & \vdots \end{bmatrix}$

In words: ith column of LM is $L(i\text{th column of } M)$.

Specifically,

$$(LM)_{ij} = \sum_{k=1}^{m} L_{ik} M_{kj}$$

Example 8.6.

$$\begin{bmatrix} 1 & 0 & 1 \\ 2 & 0 & 2 \\ 4 & 1 & 1 \\ 1 & 1 & ! \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 4 & 4 \\ 5 & 3 \\ 3 & 3 \end{bmatrix}$$

This makes sense if we consider M And L as linear transformations f, g respectively. Then $LM = g \circ f$, so the ith column of LM is the contribution of the ith coordinate of v to $g \circ f$, which is equal to $g(\text{contribution of the } i\text{th coordinate to } f)$, which is equal to $L(i\text{th column of } M)$.

Proposition 8.7. For $f : V \to W$, $g : W \to U$, $\mathcal{L}, M, N$ normal bases for V, W, U respectively.

$$\left[g \circ f\right]_{\mathcal{L}}^{N} = \left[g\right]_{M}^{N} \left[f\right]_{\mathbb{L}}^{M}$$

Proof. We compute column by column by $\mathcal{L} = (v_1, \cdots, v_n)$

$$(i\text{th column of } \left[g \circ f\right]_{\mathcal{L}}^{N} = \left[g \circ f(v_i)\right]_{N} = \left[g\right]_{M}^{N} \left[f(v_i)\right]_{M}$$

$$= \left[g\right]_M^N (i\text{th column of } \left[f\right]_{\mathcal{L}}^M$$
$$i\text{th column of } (\left[g\right]_M^N \left[f\right]_{\mathcal{L}}^M$$

So $\left[g \circ f\right]_{\mathcal{L}}^M$, $\left[g\right]_{\mathfrak{M}}^N \left[f\right]_{\mathcal{L}}^M$ have the same columns, so they are the same matrix.

□

Definition 8.8. The matrix $M \in \mathbb{F}^{n\times n}$ is **upper triangular** if $M_{ij} = 0$ for $i > j$.

Proposition 8.9. If $L, M \in \mathbb{F}^{n\times n}$ are both upper triangular, so is LM.

9 09.18.23 - Lecture #9 - Subspaces

Definition 9.1. $V \subseteq W$ is a **subspace** of $(V, \oplus, \odot)$ if $(V, \oplus \restriction (V^2), \odot \restriction (\mathbb{F} \times V))$ is a vector space.

Proposition 9.2. $V \subseteq W$ is a subspace if and only if V is nonempty and closed under $\oplus$ and $\odot$.

Example 9.3. $\operatorname{span}(A)$ for any $A \subseteq V$ is a subspace of V.

Exercise 9.1. $W \subseteq V$ is a subspace if and only if $\operatorname{span}(W) = W$.

Example 9.4. For $f : V \to W$ linear transformation, the **nullspace** or **kernel** of f for $(f) := \{v \in V : f(v) = 0\}$ is a subspace.

Example 9.5.

$$\{p \in P_n(\mathbb{R}) : \int_{-1}^{1} p = 0\}$$

Example 9.6.

$$\{\begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 : x - y + 2z = 0\}$$

Example 9.7. For $f : V \to W$ linear, the **image** $\operatorname{im}(f) = \{f(v) : v \in V\}$ is a subspace of W

Theorem 9.8 (Gaussian Elimination). Given a system of linear equations $Av = b$, we can write it as an **augmented matrix** $\left[A \mid b\right]$ and perform Gaussian elimination to find solutions.

Example 9.9. Solve the system of equations

$$3x_1 + 2x_2 + x_3 = 4$$

$$x_1 + x_3 = 2.$$

$$\begin{bmatrix} 3 & 2 & 1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix} \leftrightsquigarrow \left[\begin{array}{ccc|c} 3 & 2 & 1 & 4 \\ 1 & 0 & 1 & 2 \end{array}\right] \to \left[\begin{array}{ccc|c} 3 & 2 & 1 & 4 \\ 0 & -\frac{2}{3} & \frac{2}{3} & \frac{2}{3} \end{array}\right]$$

$$\to \left[\begin{array}{ccc|c} 3 & 2 & 1 & 4 \\ 0 & -1 & 1 & 1 \end{array}\right] \to \left[\begin{array}{ccc|c} 3 & 0 & 3 & 6 \\ 0 & -1 & 1 & 1 \end{array}\right]$$

Note that $x_1 = 1$, $x_2 = 0$, $x_3 = 1$ satisfies the system of equations.

Theorem 9.10. Let A be an $n \times p$ matrix. Then for **any** elementary row operation e, there is a unique $n \times n$ matrix E such that $EA =$ the result of performing e on A.

Proof. (Uniqueness): Suppose E exists as in the theorem statement. Consider EI_n. Then it equals by definition the row operation applied to I_n, equals E. $\square$

Exercise 9.2. Such an E exists.

Corollary 9.11. Say $A \sim B$ if and only if B can be reached from A by a sequence of elementary row operations.

If $A \sim I_n$, then there is $B \in \mathbb{R}^{n \times n}$ with $BA = I_n$.

Proof. $A \sim I_n \implies \exists E_1, \cdots, E_n$ "elementary matrices" such that $E_n \cdots E_1 A = BA = I_n$. $\square$

10 09.20.23 - Lecture #10 - Inverses

Definition 10.1. A matrix $M \in \mathbb{F}^{m\times n}$ is **invertible** if there is $M^{-1} \in \mathbb{F}^{n\times m}$ such that $(MM^{-1}) = I_{m\times m}$ and $(M^{-1}M) = I_{n\times n}$

Theorem 10.2. If M is invertible, it must be **square**.

Invertible matrices correspond to invertible linear functions.

Theorem 10.3 (Gaussian Elimination). For any matrix $A \in \mathbb{F}^{m\times n}$, there is an invertible matrix G and an upper triangular matrix U with $GA = U$.

Recall the elementary row operations; observe that these are "implemented" by invertible matrices:

1. Adding a multiple of a row to another row

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 3 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

2. Scaling a row

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

3. Swapping two rows

$$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Definition 10.4. A matrix A is in **row echelon form** if

1. A is upper triangular
2. The first nonzero entry in any row is a 1 (call this a leading 1)
3. If a_{ij} is a leading 1, then all other entries in the jth column are 0
4. The all 0 rows come at the bottom

Theorem 10.5. For any matrix A, there is an invertible matrix G with GA in **row-echelon form**.

Theorem 10.6. If A is in row echelon form, $\dim(\ker(A))$ = # columns - # leading ones.

Definition 10.7. A **matrix inverse** of A is a matrix A'such that $AA' = AA' = I_n$.

A **left inverse** of A is a matrix B such that $BA = I_n$.

A **right inverse** of A is a matrix C such that $AC = I_n$.

Exercise 10.1. Solve the system

$$\begin{cases} 3x + y \equiv 3 & \mod 11 \\ x + 2y \equiv 7 & \mod 11 \end{cases}$$

for $x, y, z \in \mathbb{Z}$.

In $(\mathbb{Z}/11\mathbb{Z})^2$,

$$\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x & y \end{bmatrix} = \begin{bmatrix} 3 & 7 \end{bmatrix},$$

so $\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}^{-1} \in (\mathbb{Z}/11\mathbb{Z})^{2\times 2}$

Since $ad - bc = 5$, and since $9 \cdot 5 = 1$, the inverse in $\mathbb{Z}/11\mathbb{Z}$ is $9 \begin{bmatrix} 2 & 10 \\ 10 & 3 \end{bmatrix} = \begin{bmatrix} 7 & 2 \\ 2 & 5 \end{bmatrix}$

So $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 7 & 2 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 7 \end{bmatrix} = \begin{bmatrix} 2 \\ 8 \end{bmatrix}$, i.e. $x \equiv 2 \pmod{11}$ and $y \equiv 8 \pmod{11}$

Exercise 10.2. Determine whether the system

$$\begin{cases} 3x + 7y + 5z \equiv 1 & \mod 11 \\ x + 2y + 3z \equiv 6 & \mod 11 \\ 5x + y + 10z = 0 & \mod 11 \end{cases}$$

has unique solutions mod 11.

11 09.22.23 - Lecture #11 - Inverses cont.

Proposition 11.1. A is surjective if and only if every system of linear equations $Ax = b$ has at least one solution if and only if A has a right inverse.

Proposition 11.2. A is injective if and only if every system of linear equations $Ax = b$ has at most one solution if and only if A has a left inverse

Proposition 11.3. If A is invertible, then any left inverse of A is a right inverse (and vice versa), and the inverse of A is unique

Proof. Suppose L is a left inverse of R is a right inverse, then $(LA)R = LAR = L(AR)$. The uniqueness of inverse is the same argument as uniqueness of additive inverses. □

One place where matrix inverses show up is when converting between bases.

Recall:

$$\begin{tikzcd} V \arrow[r, "T"] \arrow[d, "{[*]_{\mathcal{L}}}"'] & W \arrow[d, "{[*]_{\mathcal{M}}}"] \\ \mathbb{F} \arrow[r, "{[T]_{\mathcal{L}}^{\mathcal{M}}}"] & \mathbb{F} \end{tikzcd}$$

Definition 11.4. We call $[\mathrm{id}]_{\mathcal{L}}^{\mathcal{M}}$ the **change of basis matrix** from $\mathcal{L}$ to $\mathcal{M}$.

Proposition 11.5. $\left[\mathrm{id}\right]_{\mathcal{L}}^{\mathcal{M}}$ is invertible with inverse $\left[\mathrm{id}\right]_{\mathcal{M}}^{\mathcal{L}}$

Proof.

$$\left[\mathrm{id}\right]_{\mathcal{M}}^{\mathcal{L}} \left[\mathrm{id}\right]_{\mathcal{L}}^{\mathcal{M}} = \left[\mathrm{id} \circ \mathrm{id}\right]_{\mathcal{L}}^{\mathcal{L}} = \left[\mathrm{id}\right]_{\mathcal{L}}^{\mathcal{L}} = I_{n\times n}$$

□

Since invertible matrices are highly useful, we'd like to know when matrices are invertible.

Proposition 11.6. If A is invertible and $n \times n$, then A row reduces to $I_{n\times n}$.

Proof. A row reduces to some B in row eheclon form. Since A is invertible, $\dim(\ker(A)) = 0$.

So there are n leading 1s in B. That is, **every** row and column of B has a leading 1. So $B = I_{n\times n}$. □

Row reducing A is secretly multiplying GA to get A in row echelon form. Need to find G. Can apply some row operations to I as we row reduce A.

12 09.25.23 - Lecture #12 - Rank-Nullity

Definition 12.1. For $f : V \to W$,

$$\operatorname{im} f = f[V] = \{f(v) : v \in V\}$$

$$\operatorname{rank}(f) = \dim(\operatorname{im}(f))$$

$$\operatorname{null}(f) = \dim(\ker(f))$$

Proposition 12.2 (Injectivity and Surjectivity).

$$\operatorname{rank}(f) = \dim(w) \iff \forall w \in W, f(x) = w \text{ has a solution} \iff f \text{ surjective}$$

$$\operatorname{null}(f) = 0 \iff \ker(F) = \{0\} \iff f \text{ injective}$$

Proposition 12.3 (Subspaces).

$$\operatorname{im} f \subseteq W \implies 0 \leq \operatorname{rank}(f) \leq \dim(W)$$

$$\ker f \subseteq V \implies 0 \leq \operatorname{null}(f) \leq \dim(V)$$

Theorem 12.4 (Rank-Nullity Theorem). For any linear $f : V \to W$,

$$\operatorname{rank}(f) + \operatorname{null}(f) = \dim(V)$$

Proof. Let $f(v_1), \cdots, f(v_n)$ be a basis for $\operatorname{im}(f)$ with $v_1, \cdots, v_n \in V$, so $\operatorname{rank}(f) = n$, and let $u_1, \cdots, u_m$ be a basis for $\ker(f)$, so $\operatorname{null}(f) = m$.

I claim $\{v_1, \cdots, v_n, u_1, \cdots, u_m\}$ is a basis for V.

Spanning: Fix $v \in V$, say $f(v) = \alpha_1 f(v_1) + \cdots + \alpha_n f(v_n)$.

Consider $v = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n$. Then $f(v - \alpha_1 v_1 - \cdots - \alpha_n v_n) = f(v) - \alpha_1 f(v_1) - \cdots - \alpha_n f(v_n) = 0$, so $v - \alpha_1 v_1 - \cdots - \alpha_n v_n \in \ker f = \operatorname{span}(\{u_1, \cdots, u_m\}$, thus $v - \alpha_1 v_1 - \cdots - \alpha_n v_n = \beta_1 u_1 + \cdots + \beta_n u_n$.

Rearranging, we get $v = \alpha_1 v_1 + \cdots + \alpha_n v_n + \beta_1 u_1 + \cdots + \beta_n u_n \in \operatorname{span}(\{v_1, \cdots, v_n, u_1, \cdots, u_m\}$

Linearly independent: Suppose $\alpha_1 v_1 + \cdots + \alpha_n v_n + \beta_1 u_1 + \cdots + \beta_m v_m = 0$. Then $\alpha_1 f(v_1) + \cdots + \alpha_n f(v_n) + f(\beta_1 u_1 + \cdots + \beta_m u_m) = 0$. By linear independence of $f(v_1), \cdots, f(v_n)$, $\alpha_1 = \alpha_2 = \cdots = \alpha_n = 0$. This means $\beta_1 u_1 + \cdots + \beta_m u_m = 0$. By linear independence of $u_1, \cdots, u_m$, $\beta_1 = \beta_2 = \cdots = \beta_m = 0$,

So all coefficients are 0, and $\{v_1, \cdots, v_n, u_1, \cdots, u_m\}$, hence $\dim(V) = n + m = \operatorname{rank}(f) + \operatorname{null}(f)$. $\square$

Theorem 12.5 (2-out-of-3 theorem for functions). If $f : V \to W$ is linear, any two of the following three properties imply f is an isomorphism (and so also imply the third property)

1. $\dim V = \dim W$
2. f is surjective
3. f is injective

Corollary 12.6. If A is a square matrix and left invertible, then A is invertible.

Proof. The function $f : \mathbb{F}^n \to \mathbb{F}^n$, $v \mapsto Av$ is injective, so it is an isomorphism. □

Exercise 12.1. Let V, W be finite dimensional vector spaces, $\dim V = \dim W$. Then $T : V \to W$ is an isomorphism if and only if T is a linear injection

Corollary 12.7. $\varphi : V \to V^{**}$, $V \mapsto ev_v$ is an isomorphism, if $\dim V < \infty$

Exercise 12.2. Let A be an $n \times n$ matrix, and $W_m = \ker(A^m)$. Prove that the sequence $W_1, W_2, \cdots, W_n, \cdots$ eventually stabilizes, i.e. for some N, for every $i, j \geq N$, $W_i = W_j$.

Theorem 12.8. Let V be a K-vector space. Then V is infinite dimensional if and only if there is a sequence of subspaces $V_1 \subseteq V_2 \subseteq \cdots \subseteq V_n \subseteq \cdots$ which never stabilizes

Proof. Forward: As V is infinite dimensional, there is $I = \{v_n : n \in \mathbb{N}\}$ infinitely independent.

Take $V_n = \operatorname{span}\{v_1, \cdots, v_n\}$. This works!

Reverse: Take $V_1 \subset V_2 \subset \cdots \subset V_n \subset \cdots$, our sequences wlog assume inclusions are strict

Take $v_1 \in V_2 \setminus V_1, v_2 \in V_3 \setminus V_2, \cdots, v_n \in V_{n+1} \setminus V_n, \cdots$

Claim: $\{v_n : n \in \mathbb{N}\}$ is independent

Proof. $\{v_1, \cdots, v_m\}$ is independent for every $m \in \mathbb{N}$

By induction. Base: $m = 1$ easy. If holds for some $m \in \mathbb{N}$, note that if $\sum_{i=1}^{m+1} x_i v_i = 0$, then $v_{m+1} \notin \operatorname{span}\{v_1, \cdots, v_m\}$, we have $\lambda_{m+1} = 0$. But the induction of $\{v_1, \cdots, v_m\}$ yield $\lambda_1 = \cdots = \lambda_m = 0$.

To conclude, we note that if $I_0 \subseteq \{v_n : n \in \mathbb{N}\}$ is finite, then there is $m \in \mathbb{N}$ maximal such that $v_m \in I_0$. Then $I_0 \subseteq \{v_1, \cdots, v_m\}$ (which is linearly independent), so it's independent. □

□

13 09.27.23 - Lecture #13 - Rank-Nullity cont.

Example 13.1.

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

rank = 2
nullity = 1

Example 13.2.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

nullity = 0
rank = 4

Example 13.3.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

nullity = 0
rank = 4

Example 13.4.

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 & 0 \end{bmatrix}$$

rank = 4
nullity = 2

Example 13.5.

$$\begin{bmatrix} 1 & 1 & 3 & 1 \\ 0 & 2 & 2 & 5 \\ 0 & 0 & 3 & 6 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

rank = 4

nullity = 0

Example 13.6.

$$\begin{bmatrix} 1 & 1 & 3 & 1 \\ 0 & 2 & 2 & 5 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

nullity = 1
rank = 3

Exercise 13.1. If A is $n \times n$ and upper triangular, then $\text{null}(A)$ = number of 0's on the diagonal.

Note that if given an equation involving a linear function, if the number of boundary conditions is less than the rank, there will be no unique solution; if they are equal, there will be a unique solution, and if there are more boundary conditions, there would be no solutions unless these additional conditions agree with the afore unique solution.

14 09.29.23 - Lecture #14 - Column-Echelon Form

We can use column reduction to compute a basis for the span of a matrix.

Example 14.1.

$$\begin{bmatrix}1&2&6&1&3\\1&1&3&2&6\\1&2&6&1&3\\1&1&3&4&2\end{bmatrix}\to\begin{bmatrix}1&0&6&1&3\\1&-1&3&2&6\\1&0&6&1&3\\1&-1&3&4&2\end{bmatrix}\to\begin{bmatrix}1&0&0&1&3\\1&-1&0&2&6\\1&0&0&1&3\\1&-1&0&4&2\end{bmatrix}\to\begin{bmatrix}1&0&0&0&3\\1&-1&0&0&6\\1&0&0&0&3\\1&-1&0&2&2\end{bmatrix}$$

$$\to\begin{bmatrix}1&0&0&0&0\\1&-1&0&0&0\\1&0&0&0&0\\1&-1&0&2&0\end{bmatrix}$$

Hence the basis for the image is now:

$$\begin{bmatrix}1\\1\\1\\1\end{bmatrix},\begin{bmatrix}0\\-1\\0\\-1\end{bmatrix},\begin{bmatrix}0\\0\\0\\2\end{bmatrix}$$

So rank $= 3$ and nullity $= 2$

Definition 14.2. For a matrix $A \in \mathbb{F}^{m\times n}$, the **transpose** is $A^T \in \mathbb{F}^{n\times m}$ with entries

$$(A^T)_{ij} = A_{ji}.$$

Example 14.3.

$$(AB)^T = B^T A^T$$

Definition 14.4. A matrix A is in **column-echelon** form if A^T is in row-echelon form.

Proposition 14.5. If A is in column-echelon form, then the columns with leading 1s form a basis for $\operatorname{im}(A)$.

Proof. For any column of A, we have

$$(i\text{th column of } A) = Ae_i \subset \operatorname{im}(A).$$

Say $v_1, \cdots, v_n$ are the columns with leading ones. These are now linearly independent. Suppose $\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n = 0$. Say the leading one for v_i is in row j_i. Then for any other v_k, the ith entry is 0.

Taking the jth entry of the equation above, we get $\alpha_1 0 + \cdots + \alpha_{i-1} 0 + \alpha_i 1 + \alpha_{i+1} 0 + \cdots + \alpha_n 0 = \alpha_i$. So, all the coefficiants are 0.

These are also spanning. Any column without a leading 1 is 0.

Say $v_{n+1}, v_{n+2}, \cdots, v_{n+k}$ are the last k columns, all 0. Then for any $\begin{bmatrix} a_1 \\ \vdots \\ a_{n+k} \end{bmatrix} \in \mathbb{F}^{n+k}$,

$$A \begin{bmatrix} a_1 \\ \vdots \\ a_{n+k} \end{bmatrix} = a_1 v_1 + a_2 v_2 + \cdots + a_n v_n + a_{n+1} v_{n+1} + \cdots + a_{n+k} v_{n+k}$$

$$= a_1 v_1 + a_2 v_@ + \cdots + a_n v_n \in \operatorname{span}(v_1, \cdots, v_n).$$

So, $\operatorname{im}(A) \subseteq \operatorname{span}(v_1, \cdots, v_n)$.

□

Proposition 14.6. If $A \in \mathbb{F}^{m \times n}$, $G \in \mathbb{F}^{n \times n}$ is invertible, then

$$\operatorname{im} A = \operatorname{im} AG.$$

Proof. Say $v \in \operatorname{im} AG$, so $v = AGu$ for some $u \in \mathbb{F}^n$. Then $v = A(Gu)$ is in the image of A.

Suppose for $v \in \operatorname{im} A$, say $v = Au$. Set $w = Gu$, then

$$AGw = AG(G^{-1}u) = A(GG^{-1})u = Au = v,$$

so $v \in \operatorname{im}(AG)$. □

Theorem 14.7. For any matrix A there is an invertible matrix G with AG in column-echelon form.

Proof. First note that if B is invertible, then $B^T(B^{-1})^T = (B^{-1}B)^T = I^T = I$, and similarly, $(B^{-1})^T B^T = I$, so B^T is invertible.

We know there is some G invertible with GA^T in row-echelon form, so $(GA^T)^T = (A^T)^T G^T$ is in column-echelon form and G^T is invertible, and $(A^T)^T = A$. □

Given a list of vectors in $\mathbb{F}^m$, $\mathcal{L} = (v_1, \cdots, v_n)$, consider the matrix $A = \begin{bmatrix} \vdots & & \vdots \\ v_1 & \cdots & v_n \\ \vdots & & \vdots \end{bmatrix}$.

Proposition 14.8.

$$\operatorname{span}(\mathcal{L}) = \operatorname{im} A$$

Proof. $v \in \operatorname{span}(\mathcal{L})$ iff there exist $a_1, \cdots, a_n \in \mathbb{F}$ such that $v = a_1 v_1 + \cdots + a_n v_n$. This is true iff there exists some $\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \in \mathbb{F}^n$ such that $v = A \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$, which is also only true iff $v \in \operatorname{im}(A)$. □

Proposition 14.9.

(i) $\mathcal{L}$ is spanning if and only if the row-echelon of A has a leading 1 in every row

(ii) $\mathcal{L}$ is linearly independent if and only if the row-echelon form of A has a leading 1 in every column

Proof. (i) $\mathcal{L}$ is spanning if and nly if $\operatorname{im}(A) = \mathbb{F}^m$. This then holds iff $\operatorname{rank}(A) = m$, aka when the leading 1's in the row-echelon form of A is $m = \#$ of rows, which is true if and only if every row of the row-echelon form of A has a leading 1.

(ii) $\mathcal{L}$ is linearly independent if and only if for any $v \in \mathbb{F}^n$, then $Av = 0$, which implies $v = 0$. This is then true if and only if $\ker(A) = \{0\}$ and so $(A) = 0$, hence the number of leading 1s is equal to the number of columns and so every column has a leading 1.

□

Theorem 14.10. The span of multilinear alternating functions is 1-dimensional.

15 10.04.23 - Lecture #15 - Determinants

Determinants

Definition 15.1. Say $D : \mathbb{F}^{n\times n} \to \mathbb{F}$ is **multiplicative** if $D(AB) = D(A)D(B)$.

Multiplicative functions have nice properties:

Proposition 15.2. If D is multiplicative and D is not constant 0, then

1. $D(I_n) = 1$
2. If A is invertible, $D(A^{-1}) = (D(A))^{-1}$
3. D is "coordinate independent": $D([f]_{\mathcal{L}}^{\mathcal{L}}) = D([f]_{\mathcal{M}}^{\mathcal{M}})$

Proof.
1. If $D(B) \neq 0$ then $D(B) = D(IB) = D(I)D(B)$. Dividing by $D(B)$ gives $1 = D(I)$.
2. If $D(A^{-1}) \neq 0$ then $1 = D(I) = D(A^{-1}A) = D(A^{-1})D(A)$. Dividing by $D(A)$ gives $D(A^{-1}) = D(A)^{-1})$.
3. Set $Q = [\mathrm{id}]_{\mathcal{L}}^{\mathcal{M}}$. $[f]_{\mathcal{M}}^{\mathcal{L}} = [id]_{\mathcal{M}}^{\mathcal{L}}[f]_{\mathcal{M}}^{\mathcal{M}}[\mathrm{id}]_{\mathcal{L}}^{\mathcal{M}} = Q^{-1}[f]_{\mathcal{M}}^{\mathcal{M}}Q$.

   $D([f]_{\mathcal{L}}^{\mathcal{L}} = D(Q^{-1}[f]_{\mathcal{M}}^{\mathcal{M}}Q) = D([f]_{\mathcal{M}}^{\mathcal{M}})D(Q^{-1})D(Q) = D([f]_{\mathcal{M}}^{\mathcal{M}})$

□

Definition 15.3 (nonstandard). A function $D : \mathbb{F}^{n\times n} \to \mathbb{F}$ is **Gaussian-computable** if

(i) If B is gotten from A by swapping rows, then $D(B) = -D(A)$

(ii) If B is gotten from A by scaling some row by λ, then $D(B) = \lambda D(A)$

(iii) If B is gotten from A by adding a multiple of a row to another, then $D(B) = D(A)$

Proposition 15.4. If A is not invertible, then $D(A) = 0$ for every Gaussian computable D.

Proof. If A has some row that's all 0's, then scaling this row doesn't change A, so $D(A) = \lambda D(A)$ for all $\lambda \in \mathbb{F}$. This implies that $D(A) = 0$.

Otherwise, A row reduces to some matrix with an all 0's row.

Claim: If A row reduces to B then there is some $\lambda \neq 0$ with $D(A) = \lambda D(B)$.

We can check by induction on the number of steps in row reduction.

Base case: If there are 0 steps in the row reduction, then $A = B$, so $\lambda = 1$ works.

Induction step: Suppose every matrix is row reducible to B in k steps has $D(A) = \lambda D(B)$ for some λ. Suppose A row reduces to B in $k+1$ steps. Let A' be A after the first step. Then A' row reduces to B in k steps, so $D(A') = \lambda' D(B)$ for some λ'.

If the first step is scaling by r, we can let $\lambda = \frac{\lambda'}{r}$.

If it is swapping, we can take $\lambda = -\lambda'$.

If it's adding, we can take $\lambda = \lambda'$.

So if A is not invertible, $D(A) = \lambda D(B) = \lambda 0 = 0$. $\square$

Theorem 15.5. The set $\mathcal{D} = \{D : \mathbb{F}^{n\times n} \to \mathbb{F} : D \text{ is Gaussian computable}\}$ is a 1-dimension subspace of $\mathbb{F}^{\mathbb{F}^{n\times n}}$.

Definition 15.6. The determinant in dimension n is the Gaussian computable function with $\det(I_{n\times n}) = 1$.

16 10.09.23 - Lecture #16 - Determinant Properties

Theorem 16.1. The determinant is multiplicative.

Proof. Fix $B \in \mathbb{F}^{n\times n}$. Consider the map $D : \mathbb{F}^{n\times n} \to \mathbb{F}$, $A \mapsto \det(AB)$. D is multilinear and alternating. Hence for all A, $\det(AB) = D(A) = \det(B)\det(A)$ □

Theorem 16.2. $\det(A) \neq 0$ if and only if A is invertible

Proof. We showed already that if A is not invertible, $\det(A) = 0$.

If A is invertible, $\det(A)\det(A^{-1}) = \det(AA^{-1}) = \det(I) = 1$, so $\det(A) \neq 0$. □

Proposition 16.3. For $A, B \in \mathbb{F}^{n\times n}$, AB is invertible if and only if both A and B are.

Proof. AB is invertible iff $\det(AB) \neq 0$ iff $\det(A), \det(B) \neq 0$ iff both A and B are invertible. □

Theorem 16.4. $\det(A^T) = \det(A)$

Proof. Gaussian elimination says $A = E_1E_2\cdots E_nB$ with E_i elementary matrices and B in row-echelon form. Then $\det(A^T) = \det((E_1\cdots E_nB)^T) = \det(B^TE_n^T\cdots E_1^T) = \det(E_1^T)\cdots\det(B^T)$, so it suffices to check that this is true for elementary matrices and matrices in row-echelon form.

If B is not invertible, neither is B^T, so $\det(B) = 0 = \det(B^T)$.

If B is invertible and in row echelon form then $B = I_{n\times n}$ so $B^T = I_{n\times n}^T = I_{n\times n} = B$.

For swaps and scaling, $E^T = E$.

Adding one row to another: If E adds λ row i to row j. E^T adds λ row j to row i, so $\det(E) = 1 = \det(E^T)$. □

Proposition 16.5.

$$\det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$

$$\det\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = aei + bfg + cdh - gec - bfa - idb$$

Theorem 16.6. For $A \in \mathbb{F}^{n\times n}$, $B \in \mathbb{F}^{m\times m}$. Consider a block matrix $\begin{bmatrix} A & C \\ 0 & B \end{bmatrix}$, then

$$\det \begin{pmatrix} A & C \\ 0 & B \end{pmatrix} = \det(A)\det(B)$$

Theorem 16.7. If A is upper triangular, then $\det(A) = \prod_{i=1}^{n} A_{ii}$ is the product of A's diagonal entries.

Proof. By induction on n,

Base case: $\det([a_{11}]) = a_{11}$

Induction step: Suppose that $\det(A) = \prod_{i=1}^{n} A_{ii}$ when A is upper triangular. Consider B $(n+1) \times (n+1)$ upper triangular, so B is a block matrix $\begin{bmatrix} A & 0 \\ 0 & b_{(n+1)(n+1)} \end{bmatrix}$ with A upper triangular. By the above theorem, $\det(B) = \det(A)\det(b_{(n+1)(n+1)}) = \prod_{i=1}^{n} b_{ii} \cdot b_{(n+1)(n+1)}$. □

Definition 16.8. A **fixed point** of a matrix $A \in \mathbb{F}^{n\times n}$ is some $v \in \mathbb{F}^n$ such that $Av = v$.

Lemma 16.9. A has a nonzero fixed point iff $(A-I)v = Av - v = 0$ for some nonzero v iff $\ker(A-I) \neq \{0\}$ iff $A - I$ is not invertible.

So A has nonzero fixed point iff $\det(A-I) = 0$.

Definition 16.10. A map $f : K^{n\times n} \to Y$ is **coordinate-invariant** if and only if for every $A \in K^{n\times n}$, $B \in K^{n\times n}$ with B invertible, and $f(BAB^{-1}) = f(A)$.

Proposition 16.11. $f : K^{n\times n} \to Y$ is coordinate-invariant iff $f(AB) = f(BA)$ for all invertible B.

Proof. Reverse: If $f(BA) = f(AB)$ for all invertible B, then $f(B^{-1}AB) = f(ABB^{-1}) = f(A)$

Forward: $BA = B(AB)B^{-1}$. □

Definition 16.12. The **trace** is defined $\operatorname{tr}(A) = \sum_{i=1}^{n} a_{ii}$.

Corollary 16.13. Trace is coordinate-invariant.

Proof. Then

$$(AB)_{ij} = \sum_{k=1}^{n} A_{ik}B_{kj} \implies$$

$$\operatorname{tr}(AB) = \sum_{i=1}^{n}\sum_{k=1}^{n} A_{ik}B_{ki} = \sum_{k=1}^{n}\sum_{i=1} B_{ki}A_{ik} = \sum_{k=1}^{n}(BA)_{kk} = \operatorname{tr}(BA).$$

□

Theorem 16.14. Every function $f : K^{n\times n} \to K$ such that $f(AB) = f(BA)$ is a scalar multiple of the trace.

17 10.11.23 - Lecture #17 - Diagonalization

We can easily represent linear functions, systems of equations, or recurrence relations as matrices, which hypothetically allows us to manipulate them en masse and take them to large powers. However, while it's difficult to find large powers of most matrices, for diagonal matrices, we can simply raise each diagonal entry to the desired power.

Definition 17.1. v is an **eigenvector** for A if $V \neq 0$ and there exists some **eigenvalue** λ such that $Av = \lambda v$.

Definition 17.2. The **characteristic polynomial** of A is $\chi_A(x) = \det(A - xI)$

Proposition 17.3. The eigenvalues of A are the roots of χ_A.

Proposition 17.4. The eigenvectors of an eigenvalue are a basis for the kernel of $A - \lambda I$.

Definition 17.5. A basis $\mathcal{L}$ **diagonalizes** f if $[f]_{\mathcal{L}}^{\mathcal{L}} = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_n \end{bmatrix}$, equivalently, if $\mathcal{L}$ is a basis of eigenvectors. In this case, we say that f is **diagonalizable**.

How to diagonalize:

1. Find the characteristic polynomial

   e.g. $\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} = A$, then $A - xI = \begin{bmatrix} -x & 1 \\ 1 & 1-x \end{bmatrix}$

   $\det \begin{bmatrix} -x & 1 \\ 1 & 1-x \end{bmatrix} = (-x)(1-x) - 1 = x^2 - x - 2$

2. Find the roots of χ_A: $\frac{1 \pm \sqrt{5}}{2}$
3. For each eigenvalue λ, find $\ker(A - \lambda I)$

Exercise 17.1. A and B have the same eigenvalues (with multiplicity) if and only if $\chi_A = \chi_B$.

Corollary 17.6. A and A^T have the same eigenvalues.

Exercise 17.2. Prove that the following are equivalent:

(i) $A^2 = A$

(ii) A is diagonalizable and all eigenvalues are 0 or 1

18 10.13.23 - Lecture #18 - Diagonalization Guide

Definition 18.1. For an eigenvalue λ of A, the associated **eigenspace** is $E_\lambda := \ker(A - \lambda I)$.

How to diagonalize:

1. Find the eigenstuff: (eigenvalues - roots of $\chi_A(x) = \det(A - xI)$, eigenspaces - kernels of $A - \lambda I$ for each eigenvalue λ, eigenvectors - nonzero elements of eigenspaces)
2. Find B_i, a basis for ith eigenspace
3. Hope $\bigcup B_i$ is a basis for V

Note that the list of eigenvalues of A is called the spectru of A, and eigenthings are sometimes called **spectral** invariants.

This fails either when

1. A is missing eigenvalues, but this is not a problem if we work over $\mathbb{C}$ instead of $\mathbb{R}$
2. $\bigcup B_i$ fails to be a basis because it's not linearly independent
3. $\bigcup B_i$ fails to be a basis because its size is less than the dimension of V

Theorem 18.2. $\bigcup B_i$ is linearly independent.

Definition 18.3. The **geometric multiplicity** of λ is $\dim(E_\lambda) = \text{null}(A - \lambda I)$.

Definition 18.4. The **algebraic multiplicity** of λ is the number of times λ appears as a root of χ_A.

Proposition 18.5.

$$1 \leq \text{geometric multiplicity}(\lambda) \leq \text{algebraic multiplicity}(\lambda)$$

Proof. By definition, $A - \lambda I$ is not invertible, so $\text{null}(A - \lambda I) \geq 1$.

Suppose the geometric multiplicity is k, then we have $v_1, \cdots, v_k \in E_\lambda$ linearly independent vectors. Extend these to a basis $\mathcal{L}$, then $\left[A\right]_{\mathcal{L}}^{\mathcal{L}}$ is a block matrix such that the algebraic multiplicity of λ is thus greater than k (by characteristic polynomial multiplicity). $\square$

Proposition 18.6. If $A \in \mathbb{F}^{n \times n}$ and $\chi_A(x) = a_0 \prod_{i=1}^{k} (x - \lambda_i)^{m_i}$ with $\lambda_1, \cdots, \lambda_k \in \mathbb{F}$, then A is diagonalizable over $\mathbb{F}$ if and only if the geometric multiplicity and the algebraic multiplicity coincide for every eigenvalue λ.

Theorem 18.7 (Spectral Theorem). If $A \in \mathbb{R}^{n \times n}$ and $A^T = A$, then A is diagonalizable on $\mathbb{R}$.

19 10.23.23 - Lecture #19 - Inner Products

Definition 19.1. For V a real vector space, an **inner product** is a function $\langle \cdot, \cdot \rangle : V^2 \to \mathbb{R}$ that satisfies the following properties:

1. **Symmetry:** $\langle x, y \rangle = \langle y, x \rangle$
2. **Linearity in the first coordinate:** $\langle x + \alpha y, z \rangle = \langle x, \rangle + \alpha \langle y, z \rangle$
3. **Positive definite:** $\langle x, x \rangle \geq 0$ and equality holds only when $x = 0$

Note: (i) and (ii) implies that $\langle x, y + \alpha z \rangle = \langle x, y \rangle + \alpha \langle x, z \rangle$ so real inner products are **bilinear**.

Example 19.2. on $\mathbb{R}^n$

$\langle x, y \rangle_{\text{st}} = x \cdot y$ (dot product) $= \sum_{i=1}^n x_i y_i = y^T x$

1. For 1×1 matrices, $A^T = A$

$$y^T x = (y^T x)^T = x^T y^{TT} = x^T y$$

2. $z^T(x + \alpha y) = z^T x + \alpha z^T y$
3. $x^T x = \sum_{i=1}^n x_i^2 \geq 0, = 0$ only when each x_i is 0

Example 19.3. on $P(\mathbb{R})$

$\langle p, q \rangle_{\text{int}} = \int_0^1 pq$

1. commutativity of multiplication
2. $\int_0^1 (p + \alpha q) r = \int_0^1 pr + aqr = \int_0^1 pr + \alpha \int_0^1 pr$
3. $\langle p, p \rangle_{\int} = \int_0^1 p^2(x)\,\mathrm{d}x$, since $p(x) \geq 0$ for all x, $\int_0^1 p^2(x)\,\mathrm{d}x \geq 0$

Definition 19.4. For V a complex vector space, an inner product on V is a function $\langle \cdot, \cdot \rangle : V^2 \to \mathbb{C}$ that satisfies the following properties:

1. **Skew symmetry:** $\langle x, y \rangle = \overline{\langle y, x \rangle}$
2. **Linearity in the first coordinate:** $\langle x + \alpha y, z \rangle = \langle x, \rangle + \alpha \langle y, z \rangle$
3. **Positive definite:** $\langle x, x \rangle \geq 0 \in \mathbb{R}$ and equality holds only when $x = 0$

Note that (i) and (ii) imply $\langle x, y + \alpha z \rangle = \overline{\langle y + \alpha z, x \rangle} = \overline{\langle y, x \rangle + \alpha \langle z, x \rangle} = \overline{y, x} + \overline{\alpha} \overline{\langle z, x \rangle} = \langle x, y \rangle + \overline{\alpha} \langle x, z \rangle$ so over $\mathbb{C}$, inner products are linear in first slot and conjugate-linear in the second slot.

Example 19.5. On $\mathbb{C}^n$

$$\langle x, y \rangle_{\text{st}} = x \cdot y = \sum_{i=1}^n x_i \overline{y}_i = \overline{y}^T x$$

Example 19.6. On $P(\mathcal{C})$,

$$\langle p, q \rangle_{\text{int}} = \int_0^1 p\overline{q}$$

Definition 19.7. For a vector $v \in \mathbb{R}^n$, the **length** or **norm** of v is

$$\sqrt{\sum_{i=1}^{n} v_i^2} = \sqrt{\langle v, v \rangle} = \|v\|$$

Proposition 19.8. For $u, v \in \mathbb{R}^n$, by the law of cosines,

$$\cos(\theta) = \frac{|u|^2 + |v|^2 - |u - v|^2}{|u|\,|v|\,2}$$

$$\theta = \cos^{-1}\left(\frac{\langle u, v \rangle_{st}}{\|u\|\,\|v\|}\right)$$

Proposition 19.9. $u \perp v$ if and only if $\langle u, v \rangle_{st} = 0$

Theorem 19.10 (Cauchy-Schwarz inequality). For any inner product $\langle \cdot, \cdot \rangle$,

$$|\langle v, v \rangle| \leq \|u\|\,\|v\|$$

20 10.25.23 - Lecture #20 - Orthogonality

Definition 20.1. A linear bijection $T : V \to W$ is an **isometry** if $\langle Tx, Ty\rangle = \langle x, y\rangle$ for all x, y.

An isometry $T : V \to V$ is called an **orthogonal operator**.

Proposition 20.2. T is an isometry if and only if $\|T(x)\| = \|x\|$.

Example 20.3. Rotations in $\mathbb{R}^2$

$$R = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

For any v, w, we have that $\langle Rv, Rw\rangle_{\text{st}} = (Rw)^T(Rv) = w^T r^T Rv = w^T v = \langle w, v\rangle_{\text{st}}$.

Definition 20.4. For any inner product space, $A \subseteq V$ is **orthogonal** if $\langle u, v\rangle = 0$ for $u, v \in A$ distinct.

$A \subseteq V$ is **normal** if $\|v\| = 1$ for all $v \in A$.

A is **orthonormal** if A is normal and orthagonal. Equivalently, A is **orthornomal** if $\langle u, v\rangle = \delta_u(v)$ for $u, v \in A$.

Proposition 20.5. If A is orthogonal and $0 \notin A$ then A is linearly independent.

Proof. Suppose $v_1, \cdots, v_n \in A$ are distinct and $\alpha_1 v_1 + \cdots + \alpha_n v_n = 0$. Then $0 = \langle 0, v_i\rangle = \langle \alpha_1 v_1 + \cdots + \alpha_n v_n, v_i\rangle = \sum_{j=1}^n \alpha_j \langle v_j, v_i\rangle = \alpha_i \langle v_i, v_i\rangle$ Since $v_i \neq 0$ and $\langle \cdot, \cdot\rangle$ is positive definite, $\langle v_i, v_i\rangle \neq 0$, so $\alpha_i = 0$. □

Theorem 20.6. For an ordered basis $\mathcal{L}$, $[*]_{\mathcal{L}}$ is an isometry if and only if $\mathcal{L}$ is orthonormal.

Proof. The coordinate map $[\cdot]_{\mathcal{L}}$ is a linear isomorphism, so we only need to check that the inner product is preserved.

For $v = \alpha_1 v_1 + \cdots + \alpha_n v_n$ and $u = \beta_1 v_1 + \cdots + \beta_n v_n$, we have that

$$\langle v, u\rangle = \sum_{i=1}^n \sum_{j=1}^n \alpha_i \beta_j \langle v_i, v_j\rangle = \sum_{i=1}^n \sum_{j=1}^n \alpha_i \beta_j \langle e_i, e_j\rangle_{\text{st}} = \left\langle \sum_{i=1}^n \alpha_i e_i, \sum_{j=1}^n \beta_j e_j \right\rangle_{\text{st}} = \langle [v]_{\mathcal{L}}, [u]_{\mathcal{L}}\rangle_{\text{st}}$$

□

Definition 20.7. For $W \subseteq V$, define $x \perp W$ if and only if $\forall y \in W$, $\langle x, y\rangle = 0$.

Definition 20.8. For a subspace $W \subseteq V$, the **orthogonal projection** of v on W is $w = \text{proj}_W^{\perp}(v)$ such that

$$w \in W \qquad v - y \perp w \quad \forall y \in W$$

Proposition 20.9. Suppose $W = \text{span}(A)$ with A orthonormal. There are unique vectors $V_{||}$ and $v_{\perp}$ such that $v = v_{||} + v_{\perp}$ with $v_{||} \in W$ and $v_{\perp} \perp W$.

Furthermore, the orthogonal projection $\text{proj}_W^{\perp}(v) = v_{||}$ is linear.

Proof. We first show uniqueness. Suppose $v = v_{||} + v_{\perp} = u_{||} + u_{\perp}$. Then

$$0 = v_{||} - u_{||} + v_{\perp} - u_{\perp} = \Big\langle (v_{||} - u_{||}), (v_{||} - u_{||}) + (v_{\perp} - u_{\perp} \Big\rangle = \Big\langle v_{||} - u_{||}, v_{||} - u_{||} \Big\rangle,$$

hence $v_{||} - u_{||} = 0$.

We now show existence. Let $\mathcal{L} = (v_1, \cdots, v_n)$ enumerate A. Then $\text{proj}_W^{\perp}(v) = \sum_{i=1}^n \langle v, v_i \rangle v_i$. This is in W by closure, and for any j,

$$\left\langle v - \sum_{i=1}^n \langle v, v_i \rangle v_i, v_j \right\rangle = \langle v, v_j \rangle - \sum_{i=1}^n \langle v, v_i \rangle \langle v_i, v_j \rangle$$

$$= \langle v, v_j \rangle - \sum_{i=1}^n \langle v, v_i \rangle \delta_i(j) = \langle v, v_j \rangle - \langle v, v_j \rangle = 0$$

Then by linearity, $v_{\perp}$ is orthogonal to any linear combination of $v_1, \cdots, v_n$, aka every vector in W. $\square$

21 10.27.23 - Lecture #21 - Gram-Schmidt

Theorem 21.1. Every inner product space has an othonormal basis.

Proof. Suppose V has an ordered basis $\mathcal{L} = (v_1, \cdots, v_n)$. We will show by induction on i that $W_i = \text{span}(v_1, \cdots, v_i)$ has an orthonormal basis

Base case: $i = 1$: Let $W_1 = \text{span}(v_1)$; then $u_1 = v_1 / \|v_1\|$ has length 1 and spans W_1, so $\{u_1\}$ is an orthonormal basis for W_1.

Induction step: Fix i and suppose W_i has an orthonormal basis $\mathcal{L} = (u_1, \cdots, u_i)$. Set $\tilde{v}_{i+1} = v_{i+1} - \text{proj}^{\perp}_{W_i}(v_{i+1})$ (this exists because W_i has an orthonormal basis) and $u_{i+1} = \tilde{v}_{i+1} / \|\tilde{v}_{i+1}\|$. By construction, $|u_{i+1}| = 1$, and for any $j = 1, \cdots, i$, $u_j = w_i$, so $u_{i+1} \perp u_j$.

Thus $(u_1, \cdots, u_i, u_{i+1})$ is an orthonormal basis for w_{i+1}. □

The Gram-Schmidt algorithm essentially emulates this. Let $\mathcal{L} = \{v_1, \cdots, v_n\}$ be a given basis for some vector space. Then let

$$u_1 = v_1$$

$$u_2 = v_1 - \text{proj}^{\perp}_{u_1}(v_2) = v_1 - \frac{\langle v_2, u_1 \rangle}{\langle u_1, u_1 \rangle} u_1$$

$$u_k = v_k - \sum_{j=1}^{k-1} \text{proj}^{\perp}_{u_j}(v_k),$$

and if we let $e_1, \cdots, e_n$ denote the vectors $u_1, \cdots, u_n$ after normalization, $\{e_1, \cdots, e_n\}$ is now our orthonormal basis.

22 10.30.23 - Lecture #22 - Adjoints

Proposition 22.1. The following result from the fact that every vector space has an orthonormal basis.

1. Every subspace W comes with an orthogonal projection, if $\mathcal{L} = (v_1, \ldots, v_n)$ is an ordered orthonormal basis for W, $\text{proj}_W^{\perp}(v) = \langle v, v_1 \rangle v_1 + \langle v, v_2 \rangle v_2 + \cdots$

   In particular, if $\mathcal{L}$ is a basis for V, $[v]_{\mathcal{L}}$ is easy to compose: the ith entry is $\langle v, v_i \rangle$
2. Let $u, v \in V$. Then $(\forall w)\ \langle u, w \rangle = \langle v, w \rangle \iff u = v$
3. Every inner product space of dimension n is an isometry with $\mathbb{F}^n$ with the standard inner product.

We now return to the spectral theorem. As is, we have only stated it for real vector spaces; but how would we generalize this over all vector spaces?

Definition 22.2. For a linear operator $T : V \to W$, the **adjoint** of T is the linear function $T^* : W \to V$ such that

$$\langle T(v), w \rangle = \langle v, T^*(w) \rangle .$$

Proposition 22.3. Adjoints are unique and exist.

Proof. **Uniqueness:** Suppose R, S both satisfy the characterizing property. Then for any $v \in V$, $w \in W$, $\langle v, S(w) \rangle_V = \langle T(v), w \rangle_W = \langle v, R(w) \rangle_W$. So $S(w) = R(w)$ by (ii).

Existence: For a matrix $A \in \mathbb{F}^{n \times n}$ define $A^* = \overline{A^T}$ i.e. $A^*_{ij} = \overline{A}_{ji}$ (in particular, if $A \in \mathbb{R}^{n \times n}$, $A^* = A^T$). Fix orthonormal bases $\mathcal{L}$ for V and $\mathcal{M}$ for W and define T^* by $[T^*]^{\mathcal{L}}_{\mathcal{M}} = ([T]^{\mathcal{M}}_{\mathcal{L}})^*$.

We can easily check that this satisfies the characterizing property. □

Definition 22.4. An **orthogonal matrix** is $A \in \mathbb{F}^{n \times n}$ such that $A^* A = I$

Proposition 22.5.

$$A^{**} = A \qquad (AB)^* = B^* A^*$$

Proposition 22.6. For $T : V \to V$ linear and $\mathcal{L}$ an orthonormal basis, the following are equivalent:

1. T is orthogonal
2. $[T]^{\mathcal{L}}_{\mathcal{L}}$ is orthogonal
3. $T^* = T^{-1}$

Corollary 22.7. A list of vectors in $\mathbb{F}^n$, $\mathcal{L} = (v_1, \cdots, v_n)$ is an orthonormal basis if

and only if $Q = \begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix}$ is orthogonal

Definition 22.8. An operator $T : V \to V$ on an inner product space $(V, \langle \cdot, \cdot \rangle)$ is **orthogonally diagonalizable** if it has an orthonormal basis of eigenvectors.

For a matrix A, this is equivalent to $A = QDQ^*$, where Q is orthogonal and D is diagonal.

23 11.01.23 - Lecture #23 - Spectral Theorem

Definition 23.1. For the subspace $W \subseteq V$, the **orthogonal complement** of W is $W^\perp = \{v \in V : v \perp W\}$

Proposition 23.2. $W^\perp$ is a subspace.

Proposition 23.3. $W^\perp = \ker(\text{proj}_W^\perp)$

Proof. If $\mathcal{L} = (v_1, \cdots, v_n)$ is an orthonormal basis for W, then $\text{proj}_W^\perp(v) = \sum_{i=1}^n \langle v, v_i \rangle v_i$. If $v \in W^\perp$, $\langle v, v_i \rangle = 0$ for each i so $v \in \ker(\text{proj}_W^\perp)$. If $v \in \text{lcm}(\text{proj}_W^\perp)$ then $\text{proj}_W^\perp(v) = 0$, so $v = v - 0 \in W^\perp$. □

Corollary 23.4.

$$\dim(W) + \dim(W^\perp) = \dim(V)$$

Proof. $W = \text{im}(\text{proj}_W^\perp)$, so the corollary follows by Rank-Nullity. □

Corollary 23.5. $W^{\perp\perp} = W$

Proof. For $W \in W$, $v \in W^\perp$, $\langle w, v \rangle = \langle v, w \rangle = \vec{0} = 0$ so $w \in W^\perp\perp$. Thus $W \subseteq W^{\perp\perp}$. By the corollary before,

$$\dim(W) + \dim(W^\perp) = \dim(V)$$

$$\dim(W^\perp) + \dim(W^{\perp\perp}) = \dim(V),$$

so $\dim(V) = \dim(W^{\perp\perp})$ thus $W = W^{\perp\perp}$. □

Proposition 23.6. $W \cap W^\perp = \{0\}$

Proof. Suppose $v = W \cap W^\perp$. Then $\langle v, v \rangle = 0$ thus $v = 0$. □

Definition 23.7. For a subspace $W \subseteq V$, W is $T-$**invariant** if for all $w \in W$, $T(w) \in W$ (equivalently, $T[W] \subseteq W$).

Proposition 23.8. If T is self-adjoint and W is T-invariant, then $T\mid_W : W \to W$ is self-adjoint.

Proof. For $v, w \in W$, $\langle (T\mid_W)(v), w \rangle = \langle Tv, w \rangle = \langle v, Tw \rangle = \langle v, (T\mid_W)w \rangle$ □

Theorem 23.9 (Spectral Theorem). A transformation $T : V \to V$ is orthogonally diagonalizable and has only real eigenvalues if and only if it is self-adjoint ($T^* = T$).

Proof. Suppose T is self-adjoint. Say λ is an eigenvalue of T with eigenvector v. Then $Tv = \lambda v$, so $\lambda \|v\|^2 = \langle \lambda v, v\rangle = \langle Tv, v\rangle = \langle v, Tv\rangle = \langle v, \lambda v\rangle = \overline{\lambda} \|v\|^2$. Dividing by $\|v\|^2$ gives $\lambda = \overline{\lambda}$. Thus $\lambda \in \mathbb{R}$.

Now, we only need to build an orthonormal basis of eigenvectors. We proceed by induction on $\dim(V)$.

Base case: $\dim(V) = 1$: If v is any vector in V with length 1, then $\operatorname{span}(v) = V$, so $\{v\}$ is an orthonormal basis for V and $T(v) \in \operatorname{span}(V)$, so $T(v) = \lambda v$ for some v. Hence, v is an eigenvector.

Induction step: Suppose that for all V' with $\dim(V) = n$ and $T' : V' \to V'$ self-adjoint, T' has an orthonormal basis of eigenvectors, and suppose $\dim(V) = n + 1$.

Because we're working in $\mathbb{C}$, T has an eigenvector $v \neq 0$. We may assume $\|v\| = 1$. Let $V' = \operatorname{span}(v)^\perp$. Then $\dim(V') = n$, and V' is T-invariant, since for any $w \in V'$, then $\langle T(w), v\rangle = \langle w, Tv\rangle = \langle w, \lambda v\rangle = \overline{\lambda} \langle w, v\rangle = 0$.

Thus $(T\mid_{V'}) : V' \to V'$ is self-adjoint and by the induction hypothesis, $T\mid_{V'}$ has an orthonormal basis of eigenvectors, B. Eigenvectors for $(T\mid_{V'})$ are also eigenvectors for T. Because $B \cup \{v\}$ is orthonormal and $|B \cup \{v\}| = n + 1$, $B \cup \{v\}$ is an orthonormal basis of eigenvectors for T. $\square$

24 11.03.23 - Lecture #24 - Spectral Graph Theory

Definition 24.1. A **graph** is a pair $G = (V, E)$ where V is a set of vertices and E is a set of edges between them.

Write $v \sim u$ to mean that $\{u, v\} \in E$.

Proposition 24.2.

$$(A_G(f))(v) = \sum_{u \sim v} f(u)$$

When designing networks, we often want them to have few edges, because buliding connections is expensive, and we want them to be hard to disconnect.

Definition 24.3. The **degree** of a vertex is the number d of its neighbors.

Definition 24.4. A graph $G = (V, E)$ is **connected** if $\forall u, v \in V$, there exists $v_1, \cdots, v_n$ with

$$u \sim v_1 \sim \cdots \sim v_n \sim v$$

Definition 24.5. A graph G is σ-well-connected if we need to delete σk many edges to isolate a set of size k with $k < \frac{1}{2}|V|$.

Proposition 24.6. G is σ-well-connected if and only if for all $A \subseteq V$ such that $|A| < \frac{1}{2}|V|$,

$$|\{\{u, v\} \in E \mid u \in A, v \notin A\}| > \sigma |A|$$

Note that for A_G, d is an eigenvalue with an eigenvector that is the characteristic function for v, and the geometric multiplicity of d is the number of connected components.

Proposition 24.7. If f is a d-eigenvector and $|f(u)|$ is maximal, thene $f(v) = f(u)$ for $u \sim v$.

Proof. u has d neighbors, and $\sum_{v \sim u} f(v) = df(n)$. Each neighbor v has $|f(v)| \preccurlyeq |f(u)|$, then

$$\left| \sum_{v \sim u} f(v) \right| < d|f(u)|$$

□

Exercise 24.1. If λ is an eigenvalue, then $|\lambda| \leq d$.

Theorem 24.8. Say λ_2 is the second largest eigenvalue, then G is at least $\frac{d-|\lambda_2|}{2}$-well-connected.

Since computing eigenvalues of self-adjoint operators isn't too hard, this gives an efficient way of bounding how well-connected a graph is.

25 11.06.23 - Lecture #25 - Least Squares

Definition 25.1. For $T : V \to W$ and a linear equation $T(v) = b$, a **least squares solution** is v^* with $|T(v^*) - b| \leq |T(u) - b|$ for all $u \in V$.

Theorem 25.2. For $W \subseteq V$ a subspace and $b \in V$, the nearest point to b in W is $\text{proj}_W^\perp(b)$.

Proof. Let $a = \text{proj}_W^\perp(b)$. For any other $w \in W$, $b - a \perp w$ so $|b - w|^2 = |b - a|^2 + |a - w|^2$. $|a - w|^2 \geq 0$, and equality holds if and only if $w = a$. So $|b - w| \geq |b - a|$ with equality exactly when $w = a$. □

So, v^* is a least squares solution to $T(v) = b$ exactly when $T(v) = \text{proj}_{\text{im}(T)}^\perp(b)$.

Proposition 25.3.

$$\text{im}(T)^\perp = \ker(T^*)$$

Proof. $\supseteq$: Suppose $T^*(v) = 0$. Fix $u \in \text{im}(T)$, say $u = T(w)$. Then $\langle v, u\rangle = \langle v, T(w)\rangle = \langle T^*(v), w\rangle = \langle u.w\rangle = 0$.

$\subseteq$: Suppose $v \in \text{im}(T)^\perp$, then, for any $w \in V$, $\langle v, T(w)\rangle = \langle T^*(v), w\rangle = 0$. So $T^*f(v) = 0$. □

Corollary 25.4.

$$a = \text{proj}_{\text{im}\,T}^\perp(b) \iff a \in \text{im}(T) \text{ and } T^*a = T^*b$$

Proof. $T^*a = T^*b$ if and only if $a - b \in \ker(T^*) = \text{im}(T)^\perp$. Additionally, $a = \text{proj}_{\text{im}\,T}^\perp(b)$ if and only if $a \in \text{im}\,T$ and $b - a \in \text{im}(T)^\perp$ (by definition of $\text{proj}_W^\perp(b)$), if and only if $a \in \text{im}\,T$ and $T^*a = T^*b$. □

Proposition 25.5. To find a least squares solution, it suffices to solve $T^*Tv = T^*b$.

Proposition 25.6.

$$\ker(T^*T) = \ker(T)$$

Proposition 25.7. The least squares solution is unique exactly when T is injective, i.e. when T^*T is invertible.

26 11.08.23 - Lecture #26 - Minimal Norm

Definition 26.1. v^+ is a minimal norm solution to $Tv = b$ if $Tv^+ = b$ and for all $w \in V$,

$$Tw = b \implies \left|Tv^+\right| \leq |Tw|$$

Proposition 26.2. Minimal norm solutions are unique.

Proposition 26.3. The minimal norm solution to $Tv = b$ is T^*w, where w is a solution to

$$TT^*w = b.$$

Proof. Let v^+ be a minimal norm solution to $Tv = b$, then $P = \{u : Tu = b\} = \{v + u : u \in \ker(T)\}$. Since v^+ has minimal norm, we must have $v^+ \perp P$, or equivalently, $v^+ \perp \ker T$. That is, $v^+ \in \ker T^\perp = \operatorname{im} T^*$. □

A solution always exists when T is surjective, because then TT^* is invertible, so $v^+ = T^*(TT^*)^{-1}b$.

Definition 26.4. For $T : V \to W$, the pseudo-inverse of T is the function $T^\dagger : W \to V$ given by taking b to the minimal norm least squares solution to $Tv = b$.

In other words, $T^\dagger(b) = T^*w$, where w is a solution to $T^*TT^*w = T^*b$.

27 11.10.23 - Lecture#27 - Operator Norm

Question. When are two inner product spaces "close"?

Definition 27.1. Let $\{v_1, \cdots, v_n\}$ be an orthornomal basis for V.

$$\|T\|_b = \max_{ij} |\langle Tv_i, v_j\rangle|$$

$$\|T\|_2 = \sqrt{\sum_{ij} |\langle Tv_i, v_j\rangle|^2}$$

$$\|T\|_{HS} = \sqrt{\operatorname{tr}(TT^*)}$$

$$\|T\|_{op} = \max_{v\neq 0} \frac{\|T(v)\|}{\|v\|}$$

Note that the last two are basis independent, and the fourth is called the **operator norm**.

Theorem 27.2. For any $T : V \to V$ with $\dim V = n$,

$$\|T\|_{op} \leq \|T\|_2 \leq \sqrt{n}\, \|T\|_{op}$$

This implies that our choice of norm is largely irrelevant. Similar inequalities exist for any norm. We choose to focus on the operator norm because it is easy to compute for orthogonally diagonalizable operators/

Proof.

$$\|T\|_2^2 = \sum_{ij} |\langle Tv_i, v_j\rangle|^2 = \sum_{i=1}^{n} \|Tv_i\|^2 \leq n\, \|T\|_{op}^2,$$

and there is v with $\|v\| = 0$ such that

$$\|T\|_{op}^2 = \|Tv\|^2 = \sum_{i=1}^{n} |\langle Tv, v_i\rangle|^2 = \sum_{i=1}^{n}\sum_{j=1}^{n} \alpha_j^2 |\langle Tv_j, v_i\rangle|^2 \leq \|T\|_2$$

□

Proposition 27.3. Say T is orthogonally diagonalizable and has eigenvalues $\lambda_1, \cdots, \lambda_n$ with

$$|\lambda_1| \geq \cdots \geq |\lambda_n|.$$

Then $\|T\|_{op} = |\lambda_1|$.

Proposition 27.4. For a linear operator T, for any k,

$$\min\{|T - S| : \operatorname{rank}(S) \leq k\} = |T - T'|,$$

where $T'(v_i) = \begin{cases} \lambda_i v_i & i \leq k \\ 0 & \text{else} \end{cases}$

28 11.15.23 - Lecture #28 - Singular Value Decomposition

Definition 28.1. For a linear transformation $T : V \to W$ between inner product spaces, a **singular pair of bases** is a pair of orthonormal bases $\mathcal{R} = (v_1, \cdots, v_n)$ and $\mathcal{L} = (u_1, \cdots, u_m)$ such that $\langle Tv_i, u_j \rangle = 0$ for $i \neq j$.

We call $\mathcal{R}$ a basis of **right singular values** and $\mathcal{L}$ a basis of **left singular values**, and the scalars $|\langle Tv_i, u_i \rangle|$ **singular values** of T (which is independent of our choice of bases).

Definition 28.2. A matrix is **unitary** if $U^* = U^{-1}$ (same as orthogonal).

Definition 28.3. If $A \in \mathbb{C}$, a singular value decomposition of A is a factorization

$$A = U\Sigma V^*$$

where U and V are unitary matrices, and Σ is diagonal.

Note that $\mathcal{R}, \mathcal{L}$ is a singular pair of bases for A if and only if $A = \left[id\right]_{\mathcal{L}}^{st} \Sigma \left[id\right]_{st}^{\mathcal{R}}$ is a singular value decomposition and the absolute values of the entires of Σ are singular values.

Proposition 28.4. If $\mathcal{R}$ is a right singular basis for T, then $\mathcal{R}$ is a basis of eigenvectors for T^*T.

Proof.

$$\langle T^*Tv_i, v_j \rangle = \langle Tv_i, Tv_j \rangle = \langle \langle Tv_i, u_i \rangle u_i, \langle Tv_j, u_j \rangle u_j \rangle = \begin{cases} 0 & i \neq j \\ |\langle Tv_i, u_i \rangle|^2 & \text{else} \end{cases},$$

so $T^*Tv_i = |\langle Tv_i, u_i \rangle|^2 v_i$. □

Note that T^*T has positive real eigenvalues.

Corollary 28.5. The singular values of T (with respect to any singular pair of bases) are the square roots of the eigenvalues of T^*T.

Theorem 28.6. Every linear transformation between complex vector spaces has a singular pair of bases.

29 11.17.23 - Lecture #29 - SVD Applications

Proposition 29.1. The rank of A is the number of nonzero singular values.

Proof. The rank of Σ is the number of nonzero singular values, and rank doesn't change if we change the basis. □

Proposition 29.2. Say $\sigma_1, \cdots, \sigma_k$ are the only nonzero singular values, then

1. $(u_1, \cdots, u_k)$ is a basis for $\operatorname{im}(A)$
2. $(u_{k+1}, \cdots, u_n)$ is a basis for $\operatorname{im}(A)^\perp = \ker(A^*)$
3. $(v_1, \cdots, v_k)$ is a basis for $\ker(A^\perp) = \operatorname{im}(A^*)$
4. $(v_{k+1}, \cdots, v_n)$ is a basis for $\ker(A)$

Proposition 29.3. The pseudoinverse of A is

$$A^\dagger := V\Sigma^{-1}U^*.$$

Lemma 29.4. If B is unitary,

$$\|BA\|_{op} = \|A\|_{op} = \|AB\|_{op}.$$

Corollary 29.5. $\|A\|_{op}$ is the largest singular value of A.

Theorem 29.6 (Best rank r approximation). The nearest matrix to A in terms of operator norm with rank less than or equal to r is $U\Sigma' V^*$, where

$$\Sigma' = \begin{bmatrix} \sigma_1 & & & \\ & \ddots & & \\ & & \sigma_r & \\ & & & 0 \end{bmatrix}.$$

30 11.27.23 - Lecture #30

Definition 30.1. A sequence $(a_1, , a_2, a_3, \cdots) \in \mathbb{F}^{\mathbb{N}}$ satisfies a kth order linear difference equation if $\exists \beta_0, \cdots, \beta_{k-1} \in \mathbb{F}$ for all n such that

$$a_{n+k} = \beta_0 a_n + \beta_1 a_{n+1} + \cdots + \beta_{k-1} a_{n+k-1}$$

Suppose we want closed forms for recursively defined sequences; how do we find these?

Definition 30.2. The **shift map** on $\mathbb{F}^{\mathbb{N}}$ is $S : \mathbb{F}^{\mathbb{N}} \to \mathbb{F}^{\mathbb{N}}$ such that $s(a_0, a_1, \cdots) = (a_1, a_2, \cdots)$.

Remark. $(a_0, a_1, \cdots)$ *satisfies a kth order linear difference equation iff*

$$V = \operatorname{span}(\{s^n(a_0, a_1, \cdots) : n \in \mathbb{N}\}) \leq k.$$

If we can compute large powers of $s \restriction V$, we can find a closed form for $(a_0, a_1, \cdots)$.

For the example sequence $v_1 = (0, 1, 4, 12, 32, \cdots)$ and $v_2 = (1, 4, 12, 32, 80, \cdots)$ are a basis for V. Set $\mathcal{L} = (v_1, v_2)$.

If $[S^n v_1]_{\mathcal{L}} = [S^n]_{\mathcal{L}}^{\mathcal{L}} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$, then $a_n = y$.

Let's try to diagonalize $[s]_{\mathcal{L}}^{\mathcal{L}}$.

$[s]_{\mathcal{L}}^{\mathcal{L}} = \begin{bmatrix} 0 & -4 \\ 1 & 4 \end{bmatrix}$. Call this A. Then $\chi_A(x) = (x-2)^2$. Since A only has 1 eigenvalue, it cannot be diagonalized.

But consider the basis $\mathcal{M} = \left(\begin{bmatrix} -2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right)$.

Note that $(A - 2I) \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$ and that $\begin{bmatrix} -2 \\ 1 \end{bmatrix}$ is an eigenvector for A, so

$$[A]_{\mathcal{M}}^{\mathcal{M}} = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = D + N,$$

where $DN = ND$ (commutativity) and $N^2 = 0$ (nilpotent).

This makes computing $(D+N)^k$ pretty easy:

$$(D+N)^k = D^k + kD^{k-1}N + \cdots = D^k + kD^{k-1}N = \begin{bmatrix} 2^k & k2^{k-1} \\ 0 & 2^k \end{bmatrix}$$

So we can compute a closed form for the example sequence

$$[s^n(v_i)]_{\mathcal{L}} = A^n \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -2 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2^n & n2^{n-1} \\ 0 & 2^n \end{bmatrix} \begin{bmatrix} -1 & -1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2^n - n2^n \\ n2^{n-1} \end{bmatrix},$$

so $a_n = n2^{n-1}$.

Definition 30.3. A Jordan block is a matrix of the form

$$\begin{bmatrix} \lambda & 1 & & & 0 \\ & \lambda & 1 & & \\ & & \lambda & 1 & \\ & & & \ddots & 1 \\ 0 & & & & \lambda \end{bmatrix}$$

Definition 30.4. The Jordan canonical form of a matrix M (listed with multiplicity) is a matrix of the form

$$\begin{bmatrix} J_1 & & & 0 \\ & J_2 & & \\ & & \ddots & \\ 0 & & & J_n \end{bmatrix}$$

which M is similar to (and so that the eigenvalues of the blocks decrease and the size of blocks with a given eigenvalue decrease).

Theorem 30.5. Every matrix in $\mathbb{C}^{n\times n}$ has a unique Jordan canonical form.

31 11.29.23 - Lecture #31 - Jordan Canonical Form

Proposition 31.1.

1. The diagonal entries are *eigenvalues*
2. The number of blocks with diagonal entry is λ? *the geometric multiplicity of* λ
3. The sum of the lengths of these blocks is *the algebraic multiplicity of* λ

Definition 31.2. A **generalized eigenvector** of T with eigenvalue λ and **degree** n is a vector $v \in \ker(T - \lambda I)^n \setminus \ker(T - \lambda I)^{n-1}$.

Note that an eigenvector is a generalized eigenvector with degree 1.

Note that for a Jordan block of size n, e_n is a generalized eigenvector with degree n and eigenvalue λ.

Proposition 31.3. If v is a generalized eigenvector with eigenvalue λ and degree n then $\mathcal{L} = ((T - \lambda I)^{n-1}v, (T - \lambda I)^{n-2}v, \cdots, v)$ is linearly independent, $W := \operatorname{span}(\mathcal{L})$ is T-invariant, and $[T \restriction W]_{\mathcal{L}}^{\mathcal{L}}$ is a Jordan block.

Proof. To show T-invariance of W it suffices to show that $T(T - \lambda I)^i v \in W$ for each $i < n$. For $i < n - 1$, we have that

$$T(T - \lambda I)^i v = (T - \lambda I)(T - \lambda I)^i v + \lambda(T - \lambda I)^i v = (T - \lambda I)^{i+1} v + \lambda(T - \lambda I)^i v,$$

both of which are in W, and

$$T(T - \lambda I)^{n-1} v = (T - \lambda I)^n v + \lambda(T - \lambda I)^{n-1} v = \lambda(T - \lambda I^{n-1} v,$$

which is in W.

To see that $\mathcal{L}$ is linearly independent, note that $\operatorname{span}((T - \lambda I)^{n-1}v, \cdots, (T - \lambda I)^{n-k}v) \subseteq \ker(T - \lambda I)^k$, but since v has degree n as a generalized eigenvector, $(T - \lambda I)^{n-k-1}v \in \ker(T - \lambda I)^{k+1} \setminus \ker(T - \lambda I)^k$, so there are no redundant vectors.

And, by computation above,

$$[T(T - \lambda I)^{n-k} v]_{\mathcal{L}} = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \lambda \\ \vdots \\ 0 \end{bmatrix},$$

as desired. $\square$

Definition 31.4. We say a vector space V is a direct sum of $W_1, \cdots, W_n$ and write $V = W_1 \oplus W_2 \oplus \cdots \oplus W_n$ if

1. $W_i \cap W_j = \{0\}$ for $i \neq j$
2. Every $v \in V$ can be written as $v = v_1 + \cdots + v_n$ with $v_i \in W_i$.

Note that in this case the v_i's are unique.

Proposition 31.5. For $T : V \to V$ there is a basis $\mathcal{L}$ such that $[T]_{\mathcal{L}}^{\mathcal{L}}$ has a block form

$$\begin{bmatrix} B_1 & & 0 \\ & \ddots & \\ 0 & & B_n \end{bmatrix}$$

if and only if $V = W_1 \oplus \cdots \oplus W_n$ such that each W_i is T-invariant and $[T \restriction W_i]_{\mathcal{M}}^{\mathcal{M}} = B_i$ for some basis $\mathcal{M}$ of W_i.

Lemma 31.6. Fix a linear operator $S : V \to V$. Suppose $\operatorname{null}(S^n) = \operatorname{null}(S^{n+1})$ (this implies that $\ker S^k = \ker S^n$ for $k > n$), then $V = \ker S^n \oplus \operatorname{im} S^n$.

32 12.01.23 - Lecture #32 - Applications of Jordan Canonical Form

Proposition 32.1. For any A, the following expression converges

$$e^A = \frac{1}{n!}A^n$$

Proposition 32.2.

$$\begin{bmatrix} x \\ y \end{bmatrix}' = A \begin{bmatrix} x \\ y \end{bmatrix}$$

is solved by

$$\begin{bmatrix} x \\ y \end{bmatrix} = e^{At} \begin{bmatrix} x(0) \\ y(0) \end{bmatrix}$$

Corollary 32.3. The differential equation $x'' = c_1 x' + c_0 x$ has solutions of the form

$$Ae^{\lambda_2} + Be^{\lambda_1},$$

where λ_1, λ_2 are the roots of $\lambda^2 - c_1\lambda - c_0$ if these are distinct and $Ae^{\lambda} + B + e^{\lambda}$ otherwise.

Definition 32.4. A **Markov chain** is a set S of **states** and for each $x, y \in S$, a transition probability $p(x, y) \in [0, 1]$.

We think of a markov chain as a discrete time dynamical system. At time n a system can be in any of the states. Then, if it's in state x, it moves to state y at time $n + 1$ with probability $p(x, y)$.

Say a system is in state x with probability $\mathbb{P}(x)$. Then at time $n + 1$ the probability a system is in state y is

$$\sum_{x \in S} p(x, y)\mathbb{P}(x) = \mathcal{M},$$

where $\mathcal{M} : \mathbb{R}^S \to \mathbb{R}^S$ is linear.

Proposition 32.5. If $f_0 \in \mathbb{R}$ is given by $f(x) =$ the number of systems on state x at time 0, then $\mathcal{M}^n f_0(x)$ is the expected number of systems in state x at time n.

Hence, to understand systems, we need to calculate $\mathcal{M}^n$.

Theorem 32.6. Suppose $\mathcal{M}$ has eigenvalues $\lambda_1, \cdots, \lambda_n$. If $\lambda_1 = 1$, the algebraic multiplicity of λ_1 is 1, and $|\lambda_i| < 1$ for $i \neq 1$, then $\mathcal{M}$ has a limiting distribution.

Theorem 32.7 (Perron-Frobenius). If $\mathcal{M}$ is the transition matrix of a markov chain and all entries of $\mathcal{M}$ are positive, then $\mathcal{M}$ has 1 as an eigenvalue with multiplicity 1 and all other eigenvalues of $\mathcal{M}$ are smaller.

33 Recitation - Dual Systems

Definition 33.1. The **(algebraic) dual space** of a K-vector space V is the collection of linear maps $T : V \to K$, denoted V^*.

Exercise 33.1. The dual space V^* is a vector space with the operations $(\alpha T)(v) = \alpha(T(v))$ and $(T + S)(v) = T(v) + S(v)$

Exercise 33.2. $\alpha T \in V^*$

Theorem 33.2 (Dual Systems). Let V be a vector space, and let $\{v_1, \cdots, v_n\} \subseteq V$. If there are $\{f_1, \cdots, f_n\} \in V^*$ such that

$$f_i(v_j) = \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & else \end{cases},$$

then the set $\{v_1, \cdots, v_n\}$ is independent.

Proof. Suppose $\alpha_1, \cdots, \alpha_n \in K$ such that $\sum_{i=1}^n \alpha_i v_i = 0$. Let $j \in [n]$, and consider $0 = f_j(0) = f_j(\sum_{i=1}^n \alpha_i v_i) = \sum_{i=1}^n f_j(\alpha_i v_i) = \sum_{i=1}^n \alpha_i f_j(v_i) = \sum_{i=1}^n \alpha_i \delta_{ij} = \alpha_j$ □

Exercise 33.3. Let P denote the set of positive prime numbers and let $I \subseteq C^\infty(\mathbb{R}, \mathbb{R})$ (set of smooth functions, infinitely differentiable) given by $I = \{\sin(\frac{2\pi}{p}x) : p \in P\}$.

Prove that I is independent.

Lemma 33.3. Let V, W be K-vector spaces, with $S \subseteq V$ spanning, $T_1, T_2 : V \to W$ linear. If $T_1 \mid_S = T_2 \mid_S$, then $T_1 = T_2$.

Proof. Let $v \in V$ and write $v = \sum_{\alpha \in \Lambda} \lambda_\alpha v_\alpha$ where $S = \{v_\alpha : \alpha \in \Lambda\}$ and $\lambda_\alpha \in K$.

$$T_1(v) = T_1(\sum_\alpha \lambda_\alpha v_\alpha) = \sum_\alpha \lambda_\alpha T_1(v_\alpha) = \sum_\alpha \lambda_\alpha T_2(v_\alpha),$$

since they agree on every point of S.

Then by linearity,

$$= T_2(\sum_\alpha \lambda_\alpha v_\alpha) = T_2(v).$$

Since v is arbitrary, $T_1 = T_2$. □

Corollary 33.4. Let V be a K-vector space, $B \subseteq V$ be a basis. Then let $\iota : B \to V$ with $x \mapsto x$ be the inclusion map. Then for any vector space W and any function $f : B \to W$, there exists a unique $T : V \to W$ with $x \mapsto x$ linear such that $f = T \circ \iota$, or equivalently, $T \restriction_B = f$.

Proof. Suppose such a linear $T : V \to W$ exists. Then for any $v \in V$, write uniquely $v \in \sum_\alpha \lambda_\alpha b_\alpha$ where $B = \{b_\alpha : a \in \Lambda\}$. Then $T(v) = \sum_{\alpha \in \Lambda} \lambda_\alpha f(b_\alpha)$, so T is unique.

It remains to check that $T(v) = \sum_{\alpha \in \Lambda} \lambda_\alpha f(b_\alpha)$ where $v = \sum_\alpha \lambda_\alpha b_\alpha$ is a linear map. □

Theorem 33.5. Let $\{v_1, \cdots, v_n\} \subseteq V$ be linearly independent. Then there are $\{f_1, \cdots, f_n\} \subseteq V*$ such that $f_i(v_j) = \delta_{ij}$.

Proof. Let $\{v_1, \cdots, v_n\} \subseteq V$ independent. Extend I to a basis B. For each $i \in [n]$, let $g : B \to K$ via

$$g_i(v) = \begin{cases} 1 & v = v_i \\ 0 & \text{else} \end{cases}$$

Extend g_i to $f_i : V \to K$ linear in a unique way. Now $\{f_1, \cdots, f_n\} \subseteq V*$, and

$$f_i(v_j) = g_i(v_j) = \begin{cases} 1 & i = j \\ 0 & \text{else} \end{cases}.$$

□

Theorem 33.6. Suppose $\{v_1, \cdots, v_n\}$ is a basis of V. Then $\{f_1, \cdots, f_n\} \subseteq V^*$ as above are unique, *and* $\{f_1, \cdots, f_n\}$ **forms a basis** of V^*

Proof. Write $f_i = v_i^*$. Then v_i^* is characterized by $v_i^*(v_j) = \delta_{ij}$. Can we find a dual system for $B^* = \{v_i^* : i \in [n]\}$, where $B = \{v_i : i \in [n]\}$?

The maps $ev_j : V^* \to K$ given by $ev_j(f) = f(v_j)$ (evaluation map), a dual system for $\{v_i^*\}$ by above.

Exercise 33.4. Check $ev_j \in (V^*)^*$.

Let $f \in V^*$. Then f is determined by $f\mid_B$. If $\sum_{i=1}^n \lambda_i v_i^* \mid_B$, then $f(v_j) = \sum_{i=1}^n \lambda_i v_i^*(v_j) = \sum_{i=1}^n \lambda_i \delta_{ij}$

Remains to check: $f = \sum_{i=1}^n f(v_i) v_i^*$. Hence $\operatorname{span}(B^*) = V^*$; B^* is a basis equal to λ, as desired. □

Corollary 33.7. If V is finite dimensional, then $\dim V = \dim V^*$.

Exercise 33.5. $\{v_i : i \in [n]\}$ is a basis of V

Exercise 33.6. $\{v_i^* : i \in [n]\}$ is a basis of V^*

Lemma 33.8. Define $ev_i : V^* \to K$ such that $ev_i(f) = f(v_i)$, where $ev_i \in (V^*)^*$. For any $v \in V$, we have that $ev_v : V^* \to K$, $f \mapsto f(v)$ is linear (i.e. $ev_v \in V^{**}$).

Proof. Fix $v \in V$, $f, g \in V^*$, $\alpha \in K$. Then

$$ev_v(f + g) = f(v) + g(v) = ev_v(f) + ev_v(g)$$
$$ev_v(\alpha f) = (\alpha f)(v) = \alpha f(v) = \alpha ev_v(f).$$

□

Lemma 33.9. Define $\varphi : V \to V^{**}$ by $\varphi(v) = ev_v$. Then φ is linear.

Proof.

$$\varphi(v+w)(f) = ev_{v+w}(f) = f(v+w) = f(v) + f(w)$$
$$= ev_v(f) + ev_w(f) = (ev_v + ev_w)(f) = (\varphi(v) + \varphi(w))(f)$$

Also for $\alpha \in K$, $v \in V$, we have

$$\varphi(\alpha v)(f) = ev_{\alpha v}(f) = f(\alpha v) = \alpha f(v) = \alpha ev_v(f) = (\alpha\varphi(v))(f)$$

□

Lemma 33.10. If $f(v) \in 0$ for all $f \in V^*$, then $v = 0$.

Proof. Suppose $v \in V$ such that $f(v) = 0$ for all $f \in V^*$. Let $\{v_\alpha : \alpha \in \Lambda\}$ be a basis of V and expand $v = \sum_\alpha \lambda_\alpha v_\alpha$. Then $\sum_\alpha \lambda_\alpha f(v_\alpha) = 0$ for all $f \in V^*$

In particular, $\sum_\alpha \lambda_\alpha g_\beta(v_\alpha) = 0$ for all β, where g_β is defined on B via $g_\beta(v_\alpha) = \begin{cases} 1 & \alpha \in \beta \\ 0 & \text{else} \end{cases}$ extended to V by linearity.

So $0 = \sum_\alpha \lambda_\alpha g_\beta(v_\alpha) = \lambda_\beta$ So $v = \sum 0 v_\alpha = 0$. □

Lemma 33.11. φ is surjective.

Proof. Suppose $v, w \in V$ with $\varphi(v) = \varphi(w)$. Then $\varphi(v-w) = 0$, i.e. $\varphi(v-w)(f) = f(v-w) = 0$ for all $f \in V^*$. By the lemma, $v - w = 0$, so $v = w$. □

Lemma 33.12. Let $T : V \to W$ for vector spaces V and W be linear. Then $T[V]$ is a subspace of W.

Proof. $T[0] = 0$, so $0 \in T[V]$

Let $v, w \in T[v]$, then $T(u) = v$ and $T(u') = w$ for $u, u' \in V$. Then $v + w = T(u) + T(u') = T(u + u')$; since $u + u' \in V$, we also have that $v + w \in T[V]$.

Let $v \in T[V]$, $\alpha \in K$, then there exists $u \in V$ such that $v = T(u)$. But then $\alpha v = \alpha T(u) = T(\alpha u)$, so $\alpha v \in T[V]$. □

Theorem 33.13. Since φ is linear and injective and $\varphi[V]$ is a subspace of V^{**}, φ is a bijective linear map from V to $\varphi[V]$, so V^{**} has a subspace isomorphic to V.

34 Related files

Connor Gorden's Recitation A problems: `21242_A_problems.pdf`

Robert Trosten's Recitation B Problems: `21242_B_problems.pdf`

Robert Trosten's "Linear Algebra Past the Edge of Infinity": `21242_past_infinity.pdf`

Daniel Mai's Computation Examples: `21242_comp.pdf`